



Work Integrated Learning 3A.

Assignment POE Part 1 - E.SOS app: POWA (People Opposing Women Abuse).

Sheketli Mochaki ST10452756
Lerato Molokomme ST1043926
Mulweli Mudau ST10358894
Kgaogelo Mashiane ST10446936

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Information Technology in
Software Development

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1. Introduction

We have partnered with the POWA (People Oppressed Women Abuse) to develop an emergency SOS system designed to serve survivors of gender-based violence. This initiative focuses on safety, security, and accessibility, ensuring that women in crisis can connect to immediate support services. The application is built around three structured objectives: To help the women who are currently in need, establish direct connection with emergency hotlines within range, ensure inclusivity through an easy-to-navigate design that accommodates people of all abilities, including those with disabilities (POWA, 2026).

By integrating technology with POWA's holistic support services, the project advances the vision of a safe and equal society intolerant of violence, connecting them to immediate support and long-term recovery resources. By integrating technology with POWA's services, the system becomes both a lifetime and a tool for long-term empowerment. The application and development will have an estimated budget of R 309 990, which will cover the whole development cycle, justified by: app complexity, cross-platform systems, functionality, design requirements, third-party systems, and network configurations (POWA,2026).

The project team has identified the amount of potential risk including terms of late submission, overspending, cyberhacks, and third-party delays. Mitigating strategies will be implemented to minimize the challenges and safeguard project success. The successful implementation will deliver shelter to provide a safe, serene, therapeutic setting where women who have experienced abuse can start to get the help they need. In the shelters, women receive comprehensive women-centred legal services, counselling, legal advice, and skill development programs; social workers and social auxiliary workers provided by POWA provide containment and counselling to abused women around the clock; and they participate in national and regional advocacy for the defence and advancement of women's rights. (POWA,2026)

The project presents a meaningful step in towards advancing women's empowerment and equality for women and children exposed to abuse. It will not only strengthen POWA's mission but also survivors with immediate and long-term recovery resources.

Emergency SOS icon logo



1.1. Purpose of Project Charter

This project charter is beneficial because it provides clarity and alignment among all stakeholders by formally defines the project purpose, scope, and expected outcomes. In the partnership between People Oppressing Women Abuse (POWA) and the development team, it establishes a shared understand of roles, responsibilities and expectations ensuring transparency and accountability throughout the project the project lifecycle. For both organisational stakeholders, it reinforces collaboration by aligning the project with POWA'S mission and the company technical objects. It also clearly outlines the project scope and key deliverables. By documenting objectives, constraints, risks, and success criteria, the project charter serves as a reference point for decision-making, support strategic alignment with organisation goals, and increase the likelihood of project success.

2. Project And Product Overview

The POWA Emergency SOS System is aimed at developing a comprehensive mobile and web-based support platform that provides immediate assistance and access to critical support services for women and vulnerable individuals experiencing abuse, gender-based violence, domestic violence, harassment, and emotional distress. POWA (People Opposing Women Abuse) is a South African women’s rights organisation established in 1979 that provides free counselling, legal services, sheltering, advocacy, and public awareness programmes for survivors of abuse (POWA, 2026).

Gender-based violence remains a significant social challenge in South Africa, and many survivors face barriers such as fear, limited access to resources, lack of immediate transport, and delayed response times when seeking assistance. The proposed POWA’s Emergency SOS System seeks to address these challenges by introducing a centralised digital support solution that enables users to quickly access help, report incidents, and connect with the appropriate support services. This aligns with POWA’s mission of improving the quality of life of women and girls through direct services and advocacy (POWA, 2026).

The POWA’s Emergency SOS System will integrate modern emergency response functionality such as a one-tap panic button, silent alert options, real-time GPS tracking, and secure communication tools to allow survivors to discreetly request help during emergency situations. Once activated, the system will automatically transmit the user’s live location to emergency responders, trusted contacts, or POWA support staff to improve response efficiency and ensure user safety (Stark, 2025).

In addition, the system will include case reporting and evidence management features, allowing users to document incidents, upload photos, voice notes, or written reports, and store them securely for future legal or counselling purposes. A secure chat and messaging feature will allow survivors to communicate privately with social workers, counsellors, and legal advisors. The mobile application will be designed with a simple and user-friendly interface to ensure that individuals in distress can navigate it quickly and effectively, even under pressure (Stark, 2025).

From the NGO’s perspective, the POWA’s Emergency SOS System will significantly strengthen service delivery by improving communication between survivors and support teams. The system will enable POWA to receive alerts in real time, track active cases more efficiently, and respond faster to emergency situations. Through the web-based dashboard, POWA staff will be able to monitor incoming cases, manage incident reports, coordinate shelter placements, and provide follow-up support services (Stark, 2025).

Furthermore, the system will provide POWA with valuable data insights and reporting capabilities, enabling the organisation to identify trends in gender-based violence cases, high-risk locations, and common support needs. This information can support strategic decision-making, advocacy campaigns, resource allocation, and funding proposals. Beyond the organisation itself, the POWA Emergency SOS System directly benefits survivors by giving them immediate access to life-saving support, emotional counselling, legal services, and shelter resources. The application therefore serves not only as an emergency response tool but also as a long-term support and empowerment platform that contributes to women’s safety, recovery, and wellbeing (World Health Organisation, 2023).

Key Features of the POWA Emergency SOS System

Feature	Description
Mobile Application	Accessible on Android and iOS devices for 24/7 support access
One-Tap Panic Button	Allows users to send an emergency alert instantly
Silent Alert	Enables discreet requests for assistance without alerting the abuser
GPS Location Tracking	Shares real-time location with support teams and trusted contacts
Incident Reporting	Allows users to safely report abuse cases and upload evidence

Secure Messaging	Confidential communication with counsellors and support staff
Shelter and Legal Support Directory	Provides access to POWA shelters, legal aid, and counselling services
Web Dashboard	Enables POWA administrators to manage cases and monitor alerts
Awareness Resources	Educational content on abuse prevention and victim rights
Offline Functionality	Critical SOS alert capability is available in low-connectivity environments through SMS fallback mechanisms.

3. Justification

South Africa is one of the most dangerous countries in the world for women and girls. According to Statistics South Africa, a woman is killed every three hours in South Africa due to gender-based violence. The country's emergency response infrastructure, while functional in urban centers, remains fragmented and often inaccessible to women in rural, peri-urban, and economically marginalized communities. Existing digital tools are either generic, requiring users to navigate multiple menus under extreme duress, or designed without consideration for the specific needs of GBV survivors. This project addresses this critical gap (Statistics South Africa, 2023).

3.1. Business Need

POWA's existing service model relies heavily on telephonic hotlines and in-person shelter referrals. While these are effective, they depend on the survivor being able to safely make a phone call, which is a condition that is frequently impossible when an abuser is present. There is a demonstrable business need for a discreet, rapid-response digital platform that allows a survivor to call for help with a single gesture, even when they cannot speak.

The E.SOS App meets this need by providing the following capabilities:

Business Need	How E.SOS App Addresses It
Reduce emergency response time	Precise GPS data transmitted instantly to responders eliminates time wasted on locating the survivor.
Bridge communication gap during crises	One-tap SOS functionality removes the need for verbal communication, protecting users in silent-danger situations.
Scalable, affordable solution for NGOs	Cloud-based architecture allows cost-effective scaling as POWA's user base grows.
Improve personal safety outcomes	Immediate connection to trained responders and emergency services increases survival probability.
Digital transformation for POWA	Modernizes POWA's service delivery model and positions the organization as a technology-forward NGO.

3.2. Public Health and Business Impact

The Emergency SOS Services Project carries significant public health and socioeconomic implications for South Africa. Faster emergency response has a direct and measurable impact on survival outcomes in cases of physical assault, attempted femicide, and psychological crisis. Beyond individual incidents, the systemic availability of a reliable emergency platform can contribute to broader behavioral deterrence, as potential perpetrators are aware that their victims have immediate access to law enforcement (World Health Organization, 2023).

The following table summarizes the key impact dimensions:

Impact Area	Detail
Public Health	Faster emergency response directly saves lives in cases of assault, attempted femicide, and acute psychological crises. Even a 3-to-5-minute reduction in response time significantly improves medical outcomes (World Health Organization, 2023).
Crime Reduction	Real-time alerting deters criminal activity and accelerates law enforcement response, contributing to measurably safer communities in the long term (World Health Organization, 2023).
Community Wellbeing	Vulnerable populations that including women, children, elderly persons, and people with disabilities allows them gain reliable access to rapid assistance, improving overall social safety metrics (World Health Organization, 2023).
Economic Value	The platform creates a replicable model for other NGOs and municipalities, and may generate revenue through SaaS licensing, institutional subscriptions, or government partnership grants (World Health Organization, 2023).
Women's Empowerment	Providing women with a discreet, reliable safety tool increases their autonomy, confidence, and willingness to leave abusive situations, supporting POWA's long-term empowerment mandate (World Health Organization, 2023).

3.3. Strategic Alignment

The E.SOS App aligns with several key national and organizational strategic frameworks, demonstrating that the project is not merely technical exercise but a socially meaningful intervention:

Strategic Framework	Alignment
South Africa National Development Plan (NDP) 2030	The NDP explicitly calls for the elimination of gender discrimination and violence. This project directly contributes to Goal 12 (Social Protection) and Goal 13 (Building Safer Communities) (South African Government, 1997) (South African Government, 2013).
UN Sustainable Development Goals (SDGs)	Aligns with SDG 3 (Good Health and Well-Being), SDG 5 (Gender Equality), SDG 10 (Reduced Inequalities), and SDG 16 (Peace, Justice and Strong Institutions) (United Nations, 2015).
POWA Strategic Plan 2024–2027	This supports POWA's institutional goal of leveraging technology to scale its reach and improve the immediacy and quality of crisis response (POWA, 2026).
South Africa's National GBV Strategy	It complements the government's Emergency Response Action Plan (ERAP) for GBV, which prioritizes accessible, multi-channel emergency reporting mechanisms.

4. Scope

The scope of the POWA Emergency SOS System Project encompasses the complete planning, design, development, testing, implementation, and deployment of a dual-platform support system consisting of a mobile application and a web-based dashboard. The system is specifically designed to support women and vulnerable individuals experiencing abuse, gender-based violence, domestic violence, harassment, and emotional distress.

The mobile application will be developed for both Android and iOS platforms to ensure accessibility and inclusivity for a wider user base. It will provide emergency response features such as a one-tap panic button, silent alert functionality, GPS location tracking, secure messaging, incident reporting, and access to POWA support services. The web-based dashboard will enable POWA staff and administrators to monitor alerts, manage cases, coordinate shelter placements, provide counselling support, and track incident history.

The project also includes the integration of third-party services, such as real-time location services, SMS gateways, secure cloud storage, and emergency hotline connectivity. In addition, the scope covers user interface design, system security, database development, quality assurance testing, deployment, and user training. The project spans a 256-day (approximately 51-week) development cycle and is bound by the technical, financial, and human resource constraints outlined in this project charter.

4.1. Objectives

The objectives of the POWA Emergency SOS System Project are as follows:

- To develop a safe, reliable, and user-friendly emergency support application for survivors of gender-based violence and abuse.
- To provide immediate access to emergency assistance through one-tap and silent alert features.
- To improve response times by integrating real-time GPS location tracking and direct communication with support teams.
- To strengthen POWA’s ability to manage cases, monitor incidents, and provide long-term support services.
- To ensure the system is accessible to users of different abilities, including individuals living with disabilities.
- To create a platform that supports both immediate crisis intervention and long-term survivor recovery.
- To support POWA’s mission of promoting women’s safety, empowerment, and equality.

4.2. High-Level Requirements

Functional Requirements

Requirement	Description
One-Tap SOS Activation	Enables users to quickly send an emergency alert with a single tap during critical situations, ensuring immediate access to help.
Real-Time Location Sharing	Automatically shares the user’s live location with pre-selected contacts to allow accurate and fast emergency response.
Automated Emergency Notifications	Notifies emergency contacts instantly through SMS and app notifications when an SOS alert is triggered.
Operator Dashboard	Allows operators (e.g., POWA staff) to monitor, track, and respond to live SOS alerts in real time via a web interface.
Emergency Hotline Directory	Provides users with quick access to categorized emergency contact numbers such as police, ambulance, and support services.

User Account Management	Enables users to create accounts, securely log in, and manage personal details including emergency contacts and medical information.
Incident Logging	Stores all emergency events for tracking, auditing, reporting, and post-incident analysis.
Offline SOS (SMS Fallback)	Ensures SOS alerts can still be sent using SMS when there is no internet connectivity.
Multi-Language Support	Allows users to interact with the system in multiple languages to improve accessibility and inclusivity.
SAPS System Integration	Enables direct communication with official emergency services systems to improve response coordination.

Non-Functional Requirements

Requirements	Description
Fast Alert Response Time	Ensures high system performance by delivering emergency alerts almost instantly.
High System Availability	Guarantees system availability and reliability, minimizing downtime.
Data Security & Encryption	Protects sensitive user information through strong encryption standards to prevent unauthorized access.
POPIA Compliance	Ensures legal compliance with South African data protection regulations for handling personal data.
Accessibility Compliance	Ensures the application is accessible to users with disabilities through proper design standards.

4.3. Major Deliverables

Deliverable	Description
Project Charter	This foundational governance document is formally approved by all stakeholders and serves as the guiding framework for the project.
Requirements Specification Document (RSD)	Detailed functional and non-functional requirements gathered from stakeholders and end users.
System Architecture Document	Technical architecture diagrams, database schema, API specifications, and infrastructure plan.
UI/UX Design Prototypes	High-fidelity wireframes and interactive prototypes for all mobile and web screens.
Mobile Application (Android & iOS)	Fully functional, tested, and deployed mobile application accessible via app stores.
Web Operator Dashboard	Deployed and tested web portal for POWA operators to manage and respond to SOS alerts.

Security and Compliance Report	Documentation of all security measures implemented and evidence of POPIA compliance.
Test Reports	Results from unit testing, integration testing, performance testing, and user acceptance testing (UAT).
User Manual and Training Guide	End-user documentation and training materials for POWA staff and registered users.
Final Project Report	Comprehensive project closure report including lessons learned, retrospective analysis, and handover documentation.

5. SDLC Approach

The E.SOS App project adopts a hybrid Software Development Life Cycle (SDLC) approach, combining a structured, sequential model for the early governance-heavy phases with an Agile/Scrum model for development. Because the system is a safety-critical application serving survivors of gender-based violence, the team determined that Project Initiation, Requirements Analysis, and System Design required formal, gated sign-off from POWA and the academic supervisor before implementation could begin, ensuring that safety-related requirements (such as the SMS-based offline fallback and silent alert functionality) were fully understood and agreed upon before any code was written. From Development onward, the project shifts to an iterative Agile Scrum model, using two-week sprints to allow the team to demonstrate working increments, gather POWA feedback early, and adapt to changing requirements without compromising the stability of the core safety features.

SDLC Phase	Model Applied	Justification
Initiation	Sequential (Waterfall-style gate)	Formal charter approval and stakeholder sign-off are required before scope and budget can be committed.
Requirements Analysis	Sequential (Waterfall-style gate)	Safety-critical requirements must be fully elicited, reviewed, and formally signed off by POWA before design begins.
System Design	Sequential (Waterfall-style gate)	Architecture, database, and UI/UX designs must be stable and approved before development sprints can be reliably planned.
Development	Agile (Scrum, 11 x two-week sprints)	Iterative delivery allows early demonstration of core SOS functionality (Sprint 1) and incremental feature growth, with continuous POWA feedback.
Testing	Agile (Scrum, with dedicated hardening sprints)	Continuous testing within sprints is supplemented by a dedicated testing phase for full regression, security, and accessibility validation.
Deployment and Closure	Sequential (Waterfall-style gate)	Production go-live, training, and handover require a controlled, single-pass rollout rather than iterative release.

6. DevOps Lifecycle

To support the Agile development sprints, the E.SOS App project follows a DevOps lifecycle that automates the flow of code from development through to deployment, reducing manual error and enabling frequent, reliable releases. The team uses GitHub for source control and GitHub Actions for continuous integration and continuous deployment (CI/CD), Docker containers to standardise development and testing environments, and AWS Elastic Beanstalk / Azure App Services for auto-scaling production hosting. The DevOps lifecycle is organised into the following continuous stages:

Stage	Tools / Practices	Purpose
Plan	Trello / GitHub Projects, sprint backlog	Break down requirements into sprint-sized user stories and track progress against the WBS.
Code	GitHub (feature branches, Gitflow), VS Code / Android Studio	Developers work on isolated feature branches, merged via pull request and peer code review.
Build	GitHub Actions, npm / Gradle build scripts	Automatically compiles and packages the mobile app, web dashboard, and backend API on every push.
Test	Jest / JUnit unit tests, Postman/Newman API tests, GitHub Actions CI	Automated unit, integration, and API tests run on every pull request before merge is permitted.
Release	Semantic versioning, GitHub Releases	Tested builds are tagged and packaged as release candidates ready for deployment.
Deploy	Docker, AWS Elastic Beanstalk / Azure App Services, Google Play Console, App Store Connect	Automated deployment of the backend and web dashboard to cloud infrastructure; mobile builds submitted to app stores.
Operate	AWS CloudWatch, Firebase Crashlytics	Monitors uptime, error rates, and performance of the live system in production.
Monitor / Feedback	Firebase Analytics, POWA operator feedback, incident logs	Usage data and operator feedback feed back into the sprint backlog for continuous improvement.

This DevOps pipeline directly supports the project's non-functional requirements for high system availability and fast alert response time, since automated testing and staged deployment reduce the risk of introducing defects into a system that survivors depend on during emergencies.

7. Group Deliverables Allocation

The E.SOS App project deliverables are allocated across the four team members according to their designated roles, as reflected in the RACI matrix (Section 17.1, Roles and Responsibilities) and the hourly cost breakdown in the Economic Feasibility Issues table (Section 16.2). Each member is accountable for a primary deliverable stream, while contributing in a consulted or informed capacity to other streams as required.

7.1. Sheketli Mochaki - Project Manager

Sheketli Mochaki is responsible for Project Planning and Scheduling, Stakeholder Communication, POWA Liaison, Budget Tracking, and Project Closure/Handover. Deliverables include the Project Charter, the project schedule and Gantt/PERT charts, the risk register, weekly progress reports, and the final project closure report. Per the Economic Feasibility breakdown, project management activities are allocated approximately 212 hours at R200/hour, with additional hours dedicated to charter development, risk register maintenance, weekly reporting, and stakeholder communication management. Sheketli also holds Accountable/Responsible status for Security and Compliance under the Work Breakdown Structure.

7.2. Lerato Molokomme - Business Analyst

Lerato Molokomme is responsible for Requirements Gathering and Stakeholder Identification, and shares responsibility for Documentation. Deliverables include the Requirements Specification Document (RSD), the stakeholder register, use-case models, and end-user/operator training manuals. Lerato is also Accountable/Responsible for Backend and API Development under the Work Breakdown Structure, covering server and database setup, RESTful API development, user authentication and authorisation, real-time alert processing, and third-party API integration, allocated approximately 1,152 hours at R200/hour.

7.3. Mulweli Mudau - Lead Developer

Mulweli Mudau is Accountable/Responsible for System Architecture Design, Database Design, Backend/API Development support, and Security Implementation. Deliverables include the System Architecture Document, the database schema and entity relationship diagram, the class diagram, and the Security and Compliance Report. Mulweli also leads Mobile Application Development, allocated

approximately 1,200 hours at R546/hour, covering development environment setup, user registration and profile management, the one-tap SOS button, GPS and real-time location functionality, SMS/email notification integration, and incident history and reporting.

7.4. Kgaogelo Mashiane - UI/UX Designer and QA Engineer

Kgaogelo Mashiane is Accountable/Responsible for UI/UX Design, Web Dashboard Development, and Testing and Quality Assurance. Deliverables include the UI/UX design prototypes and wireframes, the interactive prototype, the Web Operator Dashboard, and the full suite of test reports (unit, integration, performance, accessibility, and user acceptance testing). Design and prototyping work is allocated approximately 480 hours at R236/hour, with Web Operator Dashboard development allocated a further 960 hours at R400/hour.

Team Member	Primary Deliverable Stream	Approx. Allocated Hours
Sheketli Mochaki	Project Management, Charter, Risk Register, Security & Compliance	212+ hours (R200/hr)
Lerato Molokomme	Requirements Analysis, Backend & API Development, Documentation	1,152+ hours (R156-R200/hr)
Mulweli Mudau	System Architecture, Database Design, Mobile App Development, Security	1,200+ hours (R236-R546/hr)
Kgaogelo Mashiane	UI/UX Design, Web Dashboard Development, Testing & QA	960+ hours (R236-R400/hr)

8. Site Map

8.1. Website Sitemap

The POWA website (the organisation's public-facing site, which will link through to the E.SOS App) follows a flat, single-level navigation structure so that a person in distress can reach help within one or two clicks. The primary navigation bar and page hierarchy are as follows:

- Home - hero message ("You Are Not Alone"), mission statement, Get Help Now and Our Mission call-to-action buttons, and an always-visible Quick Exit button.
- About POWA - organisational background, history (established 1979), and mission.
- Get Help - core support pages, reached from the main navigation and from the homepage Get Help Now button:
 - Know Your Rights
 - Legal Aid / Protection Orders
 - Signs of Abuse
 - Safe Shelters
 - Downloadable Safety Guide (PDF)
- Our App - the POWA HELP mobile app landing page, describing the Panic Button, GPS tracking, and counsellor access, with App Store and Google Play download links.
- Resources - educational and awareness content on gender-based violence.
- Donate - fundraising page ("Rebuild Lives") with Donate Now and Volunteer call-to-action buttons.
- Global elements present on every page - the site header with Donate Now button, a persistent Quick Exit safety control, and a footer containing organisational contact details, social links, and the office line.

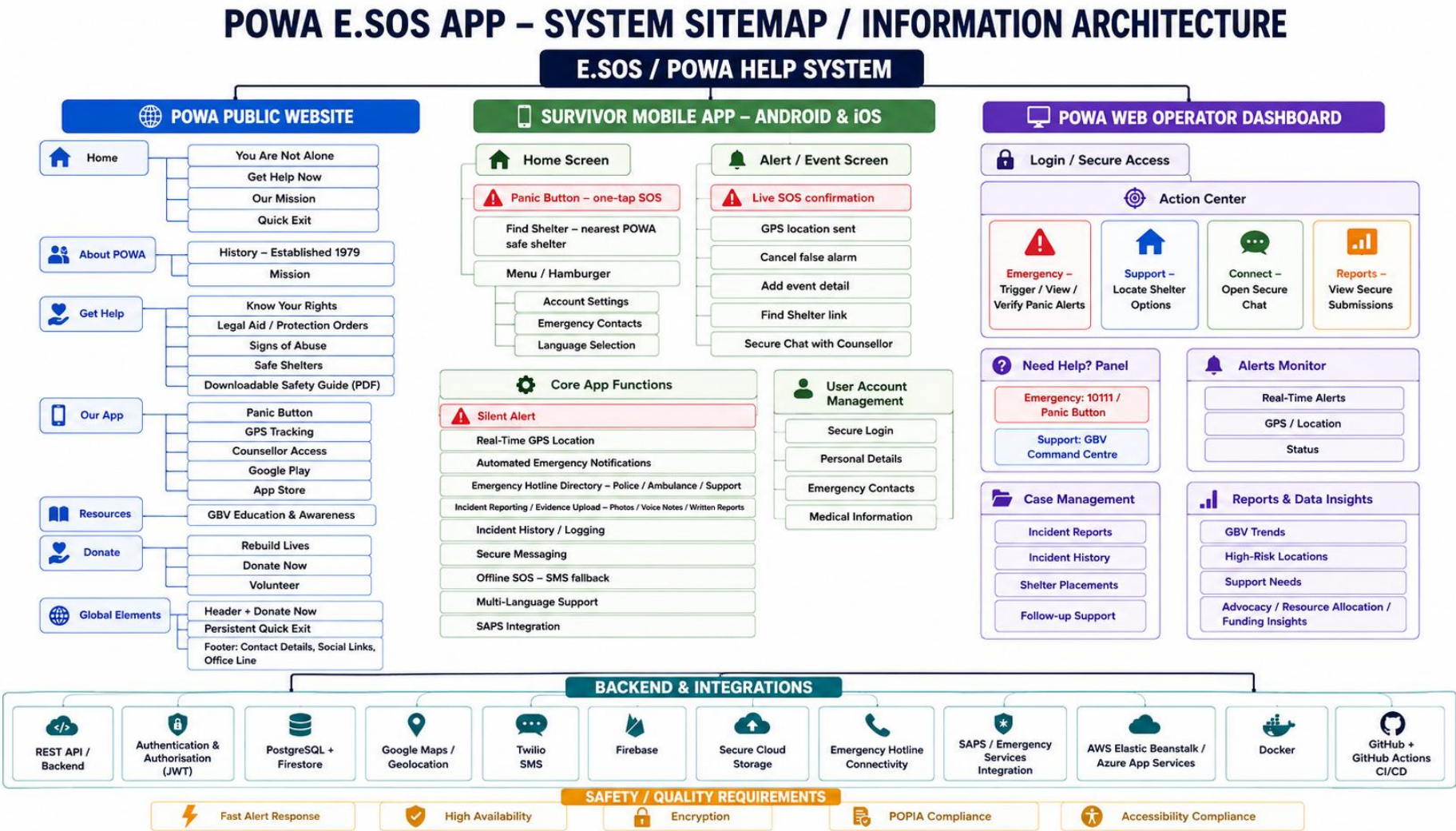
8.2. Mobile App Sitemap

The POWA HELP mobile application is structured around a minimal home screen so that the two most safety-critical actions are reachable instantly, with secondary functions accessible through the menu. The web-based Operator Dashboard used by POWA staff mirrors the same core action groups. The app and dashboard structure is as follows:

- Mobile App - Home Screen

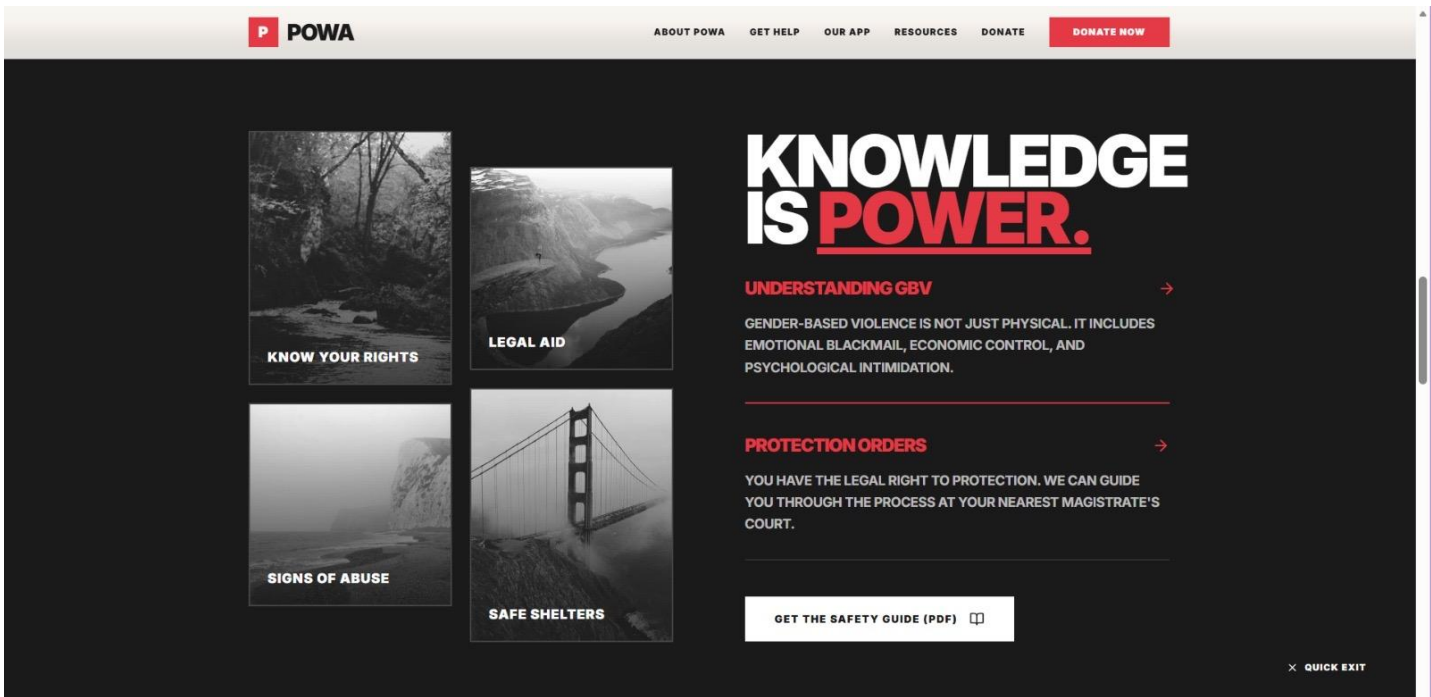
- Panic Button - triggers an immediate, one-tap emergency alert.
- Find Shelter - locates the nearest available POWA safe shelter.
- Menu (hamburger icon) - access to account settings, emergency contacts, and language selection.
- Mobile App - Alert / Event Screen (post-activation)
 - Live confirmation that the SOS alert and GPS location have been sent.
 - Option to cancel a false alarm or add further detail to the event.
 - Direct link back to Find Shelter and to secure chat with a counsellor.
- Web Operator Dashboard - Action Center
 - Trigger Panic Button (view/verify user-triggered alerts) - Emergency
 - Locate Shelter Options - Support
 - Open Secure Chat - Connect
 - View Secure Submissions - Reports
 - Need Help? panel - quick-reference Emergency (call 10111 or use the panic button) and Support (contact the GBV Command Centre) shortcuts for operators.

The reference screens below, captured from the POWA website and the POWA HELP application concept, illustrate the structure summarised above and were used to guide the wireframes in the following section.

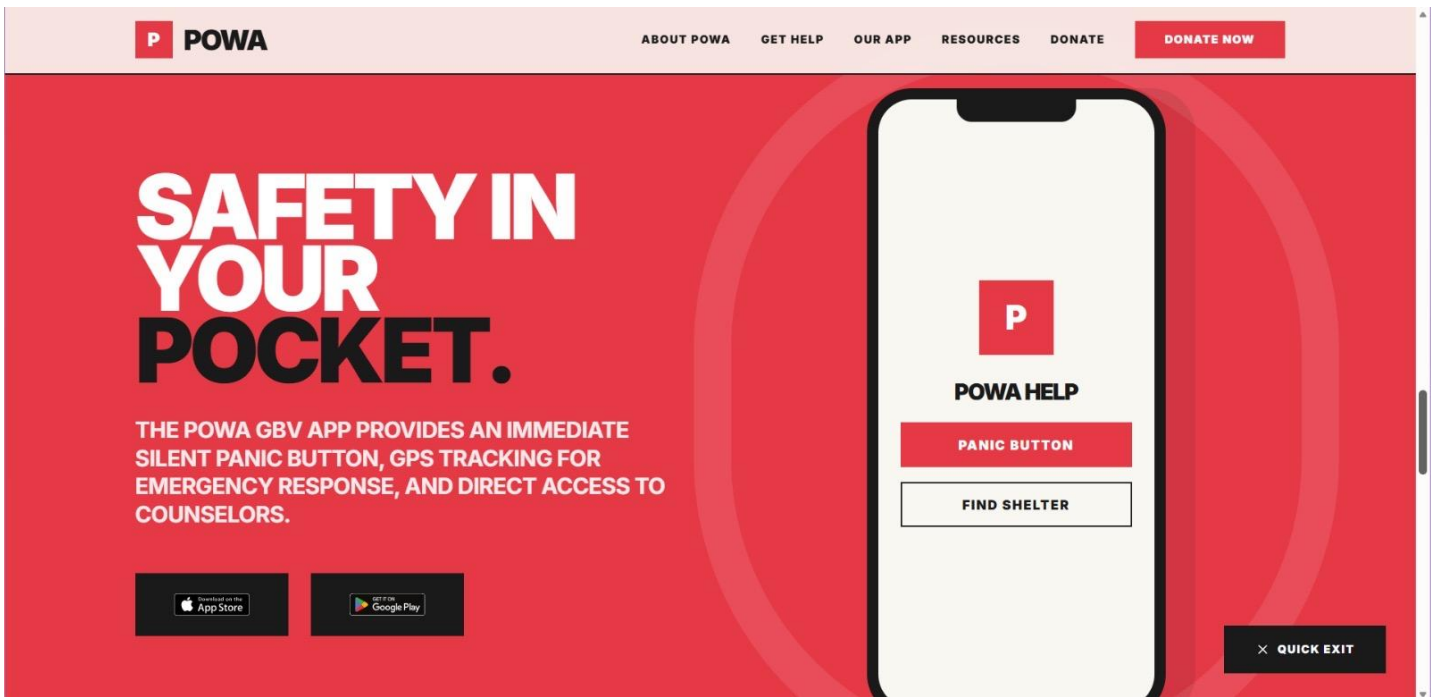


9. Wireframes

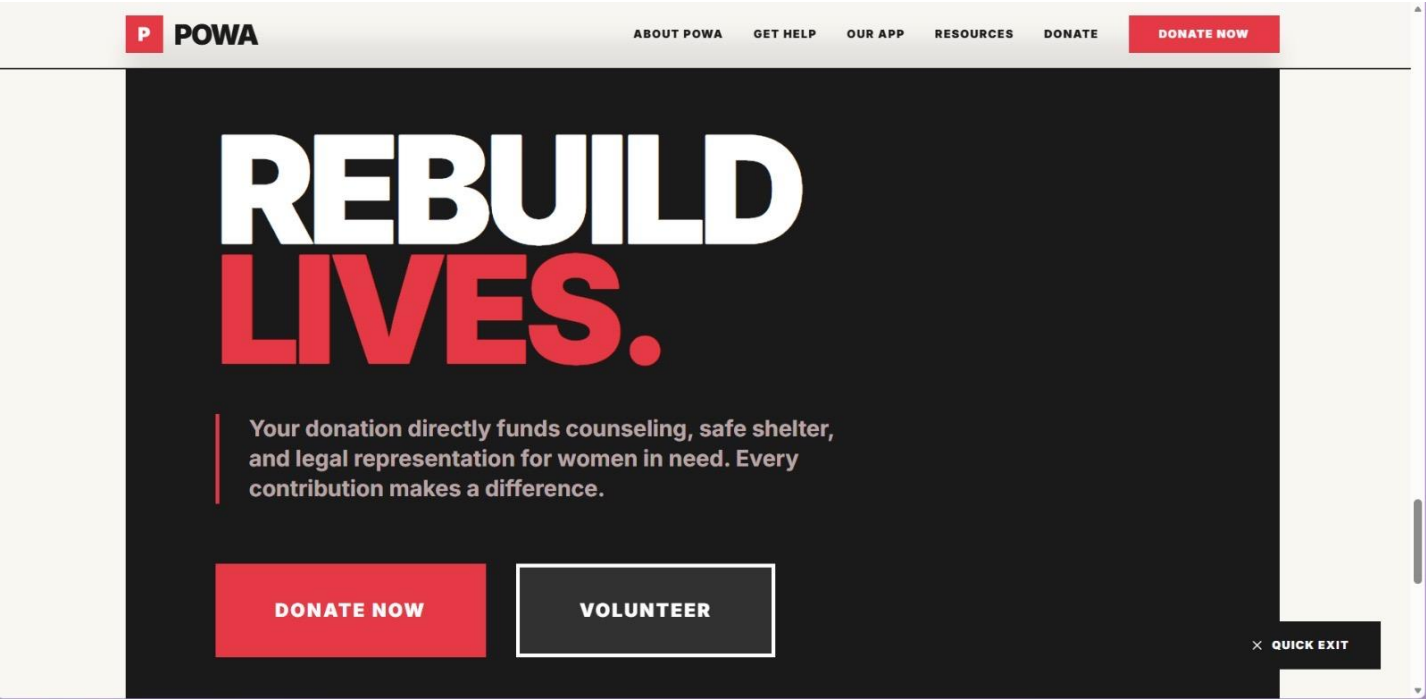
The following wireframes and high-fidelity design mockups represent the intended layout and interaction design for the E.SOS App's operator-facing web dashboard and survivor-facing mobile application. These were produced during the System Design phase and used to guide frontend implementation and stakeholder review with POWA.



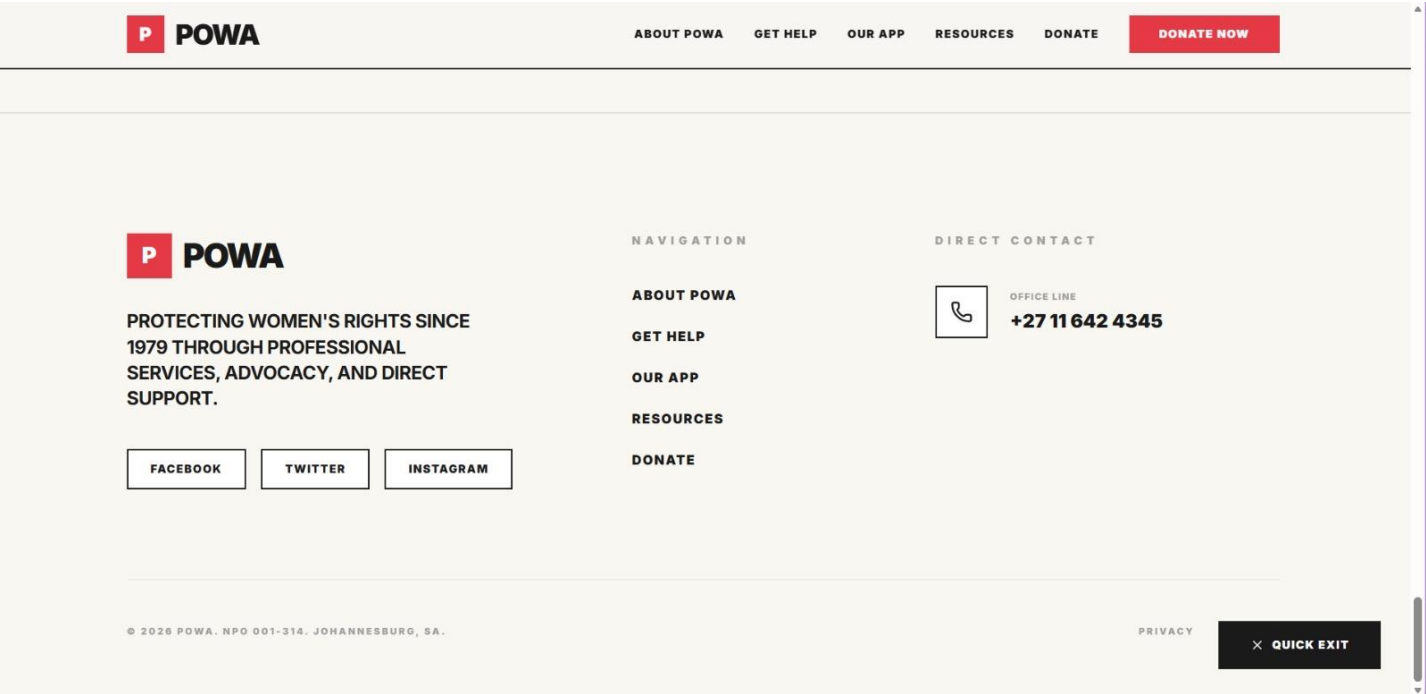
Reference screen - Website "Knowledge is Power" resource page (Know Your Rights, Legal Aid, Signs of Abuse, Safe Shelters)



Reference screen - Website "Our App" landing page (Safety in Your Pocket)



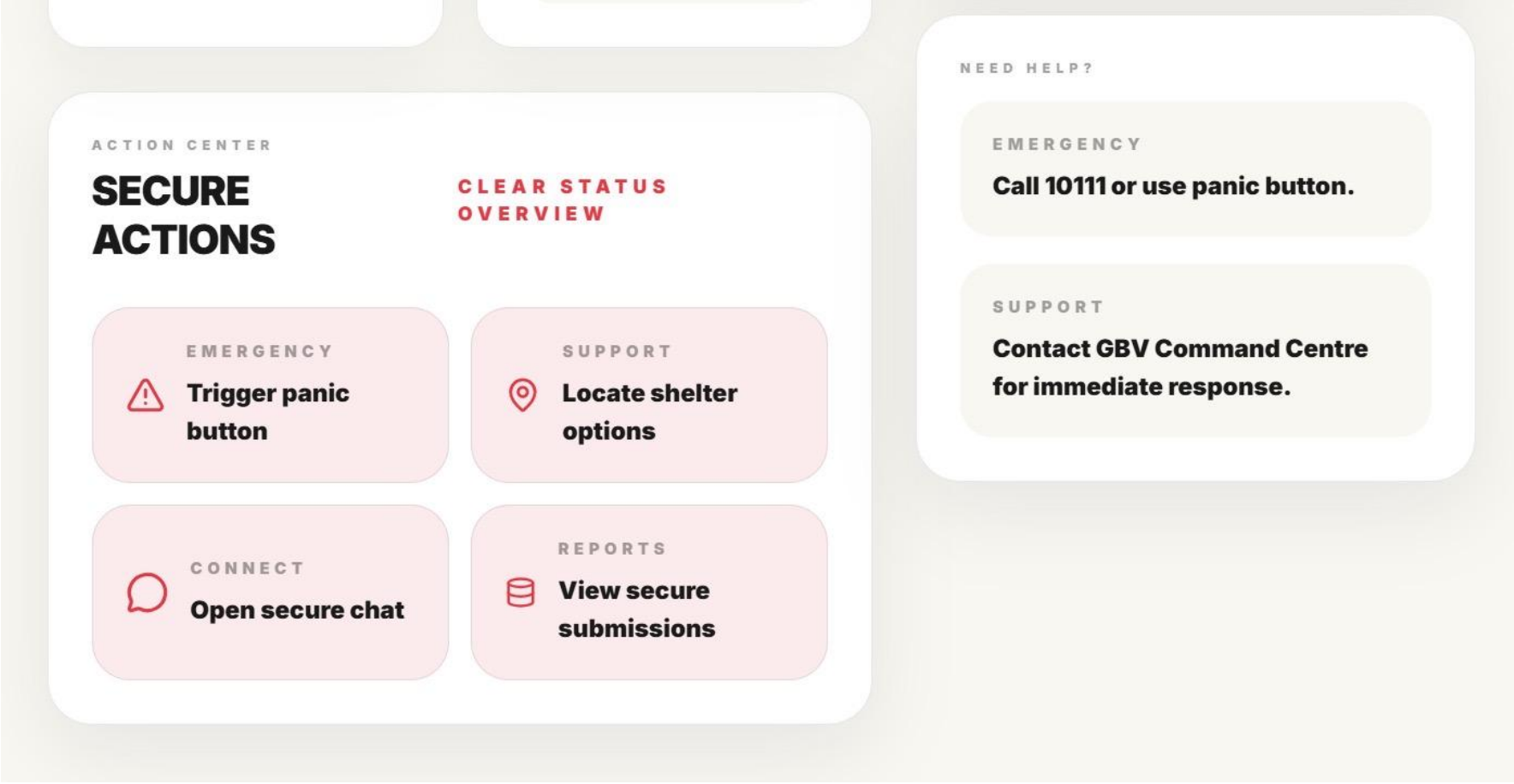
Reference screen - Website Donate page (Rebuild Lives)



Reference screen - Website footer showing global navigation and direct contact details

9.1. Website Wireframe - Dashboard

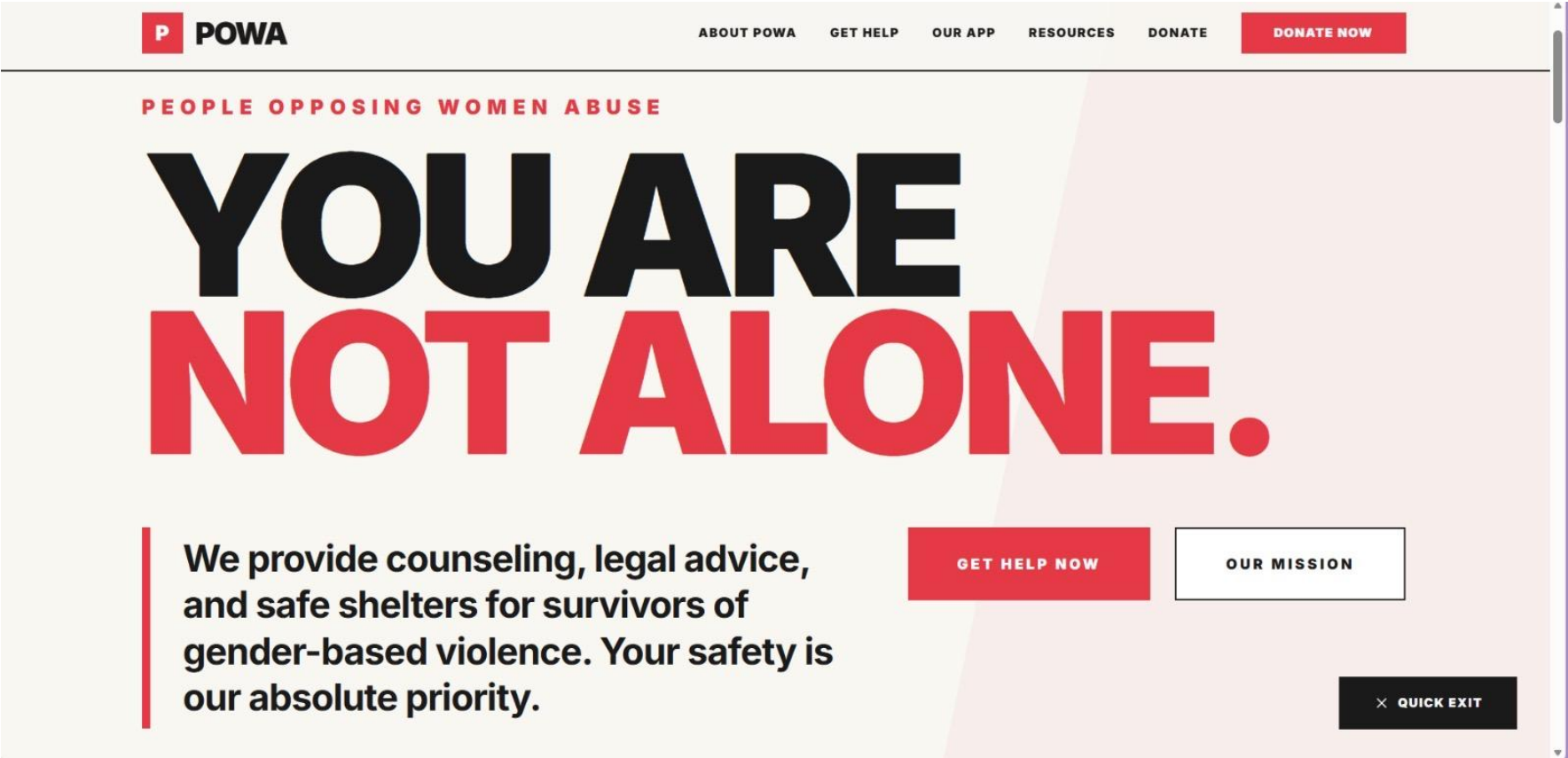
The Operator Dashboard's Action Center gives POWA staff a single screen from which to respond to an alert. The four primary actions - Trigger Panic Button (Emergency), Locate Shelter Options (Support), Open Secure Chat (Connect), and View Secure Submissions (Reports) - are presented as large, high-contrast cards for fast recognition under pressure. A persistent "Need Help?" panel on the right keeps the emergency number (10111) and the GBV Command Centre contact visible at all times.



Website Wireframe - Operator Dashboard "Secure Actions" Action Center

9.2. Website Wireframe - Home Page

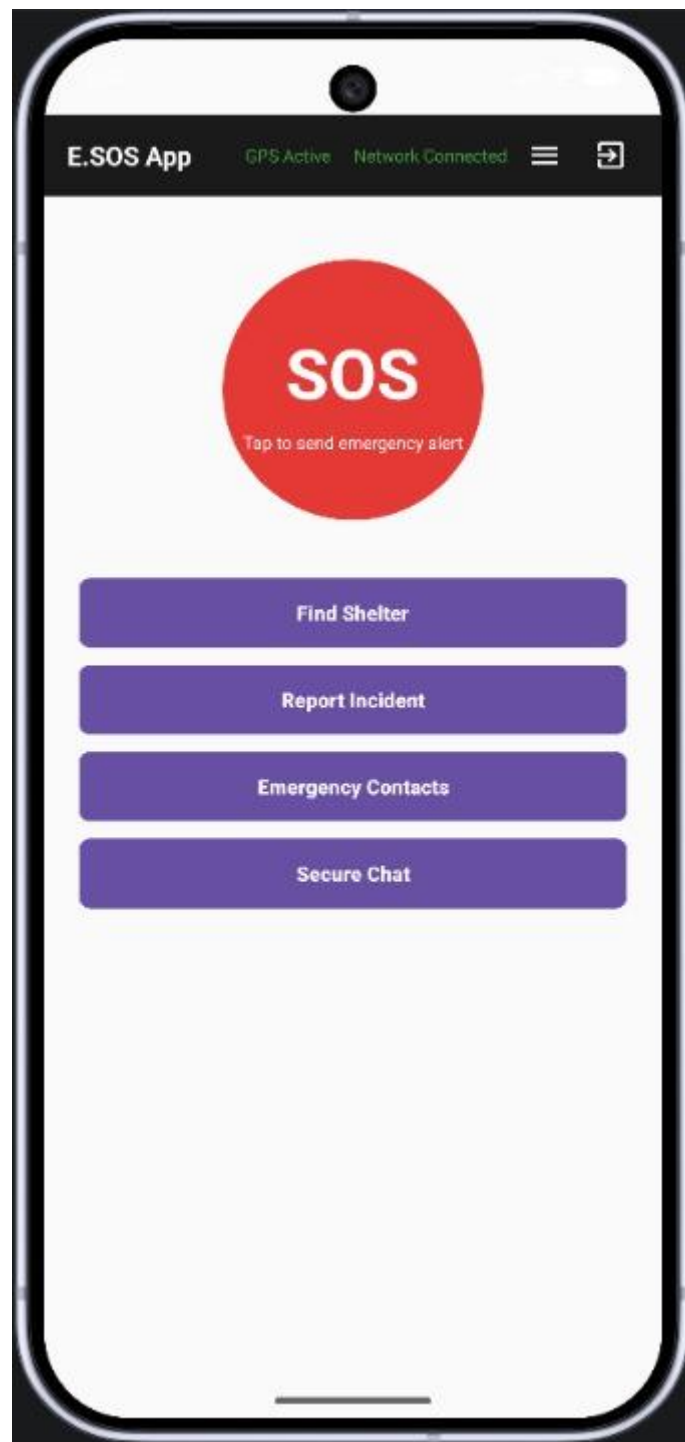
The public website home page leads with a direct, reassuring message ("You Are Not Alone") and immediately surfaces the two actions a visitor is most likely to need: Get Help Now and Our Mission. A Quick Exit control remains visible in the bottom corner throughout the site so a survivor can leave the page instantly if needed.



Website Wireframe - Home Page hero section

9.3. Mobile App Wireframe - Home Screen

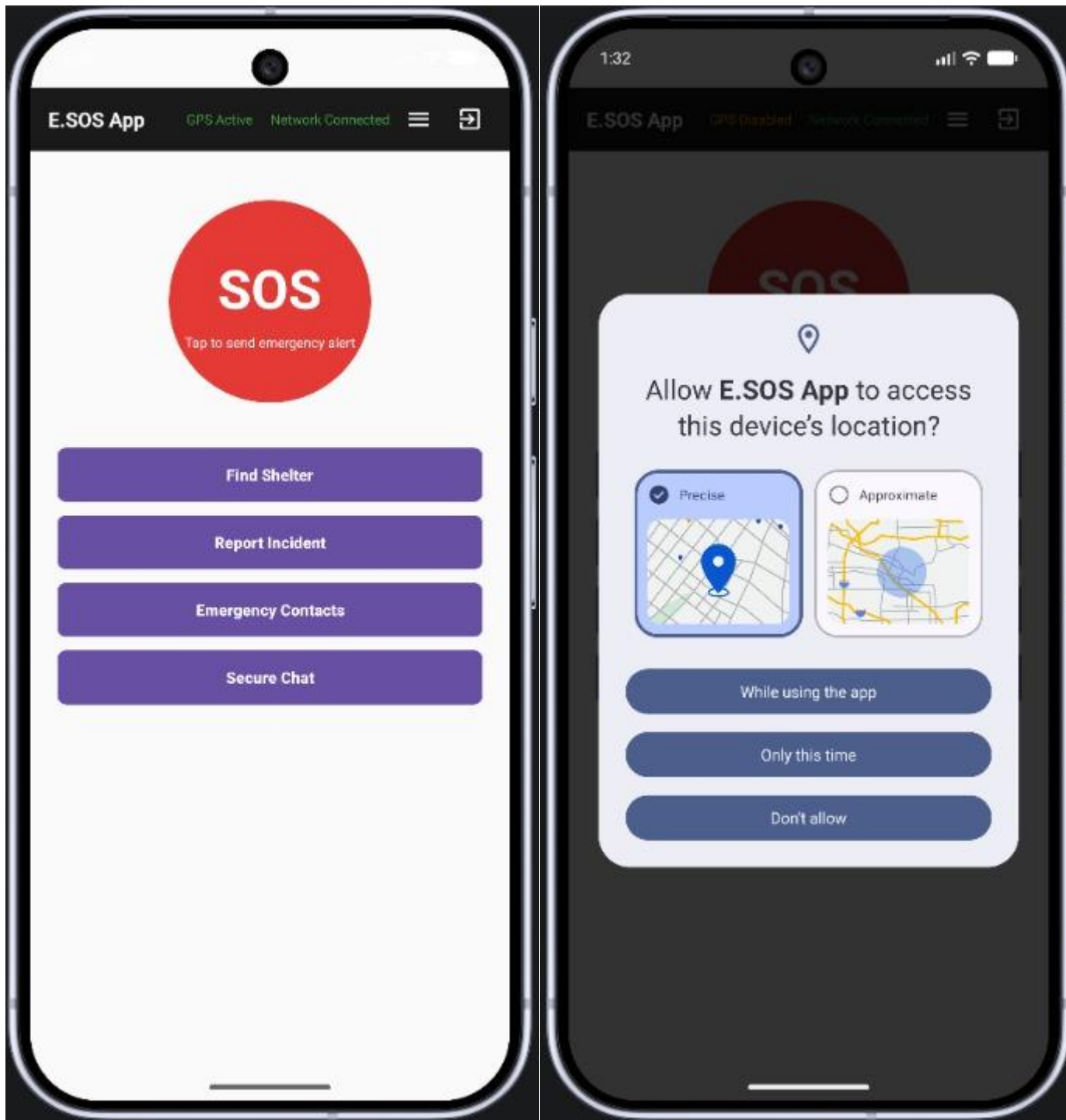
The POWA HELP mobile app opens directly onto its two safety-critical actions, Panic Button and Find Shelter, with no login wall or additional navigation required before they are reachable. This minimal first screen reflects the requirement that a distressed user must be able to act within seconds of opening the app.



Mobile App Wireframe - Home Screen

9.4. Mobile App Wireframe - Event Details

Once a user scrolls into or activates the panic flow, the Event Details screen confirms which action is about to be triggered and keeps the Panic Button and Find Shelter options clearly separated and unambiguous, reducing the risk of a mis-tap during a high-stress moment.



Mobile App Wireframe - Panic Button / Event Details screen

10. Repository and Folder Structure

The E.SOS App codebase is hosted on GitHub as a single organisation with three linked repositories, mirroring the project's three deployable components: the mobile application, the web operator dashboard, and the backend API. Each repository follows a Gitflow-inspired branching strategy (main, develop, and short-lived feature/bugfix branches), with pull requests required to merge into develop and a protected main branch that only accepts merges from tested release branches. The high-level folder structure for each repository is as follows:

- esos-mobile-app/ (React Native)
 - src/screens/ - Home, PanicButton, FindShelter, IncidentReport, SecureChat, Settings
 - src/components/ - reusable UI components (buttons, alert banners, map view)
 - src/services/ - API client, GPS/location service, SMS fallback service
 - src/navigation/ - app navigation stack and routing
 - src/assets/ - icons, images, fonts
 - __tests__/ - unit and component tests
 - android/ and ios/ - native platform build configuration

- esos-web-dashboard/ (React.js)
 - src/pages/ - Login, ActionCenter, AlertsMonitor, CaseManagement, Reports
 - src/components/ - dashboard cards, alert list, secure chat window
 - src/services/ - API client, authentication service
 - src/styles/ - shared design tokens and theming
 - tests/ - unit and integration tests
- esos-backend-api/ (Node.js / Express)
 - src/routes/ - auth, alerts, incidents, users, messages
 - src/controllers/ - request handling logic per route
 - src/models/ - database models/schema (PostgreSQL and Firestore)
 - src/middleware/ - authentication (JWT), validation, rate limiting
 - src/services/ - Twilio SMS, Google Maps Geolocation, Firebase integration
 - tests/ - unit and integration tests
 - .github/workflows/ - CI/CD pipeline definitions (build, test, deploy)
- docs/ (shared repository) - Project Charter, Requirements Specification Document, System Architecture Document, API specifications, and user manuals

11. Duration

11.1. Timeline

The Emergency SOS Services Project is planned over a structured 256-day (approximately 51-week) development cycle, divided into six distinct phases. Each phase has defined entry and exit criteria to ensure disciplined progression through the project lifecycle. The timeline below reflects a project commencement date of 02 March 2026 and a scheduled closure date of 12 February 2027. The extended timeline provides adequate buffer for thorough testing, iterative development cycles, stakeholder feedback incorporation, and high-quality documentation that are all essential for a safety-critical application serving vulnerable GBV survivors.

Phases	Phase Name	Duration	Description
Phase 1	Project Initiation	Weeks 1–3 (02 Mar – 20 Mar 2026) 15 days	Project charter development and approval, team formation, role assignment, stakeholder identification and registration, POWA partnership agreement, and formal project kickoff.
Phase 2	Requirements Analysis	Weeks 4–8 (23 Mar – 24 Apr 2026) 25 days	Comprehensive stakeholder interviews, requirements elicitation workshops, risk identification, use case modelling, and completion of the Requirements Specification Document (RSD). Includes two rounds of stakeholder review and sign-off.
Phase 3	System Design	Weeks 9–14 (27 Apr – 05 Jun 2026) 30 days	System architecture design, database schema, API contract specification, UI/UX wireframing and high-fidelity prototyping, accessibility review, security architecture planning, and formal design review and approval.

Phase 4	Development	Weeks 15–36 (08 Jun – 06 Nov 2026) 110 days	Agile sprint-based frontend, backend, and API development across 11 two-week sprints. Includes mobile app development (Android and iOS), web operator dashboard, third-party integrations, security implementation, and continuous integration pipeline setup.
Phase 5	Testing	Weeks 37–44 (09 Nov – 31 Dec 2026) 40 days	Unit testing, integration testing, performance and load testing, security penetration testing, accessibility testing, and comprehensive user acceptance testing (UAT) with POWA staff and representative end users. Includes bug fixing and regression cycles.
Phase 6	Deployment and Closure	Weeks 45–51 (05 Jan – 12 Feb 2027) 36 days	Production deployment, post-deployment monitoring, user training sessions for POWA operators, final documentation, project closure report compilation, lessons-learned retrospective, and formal handover to POWA with all source code and credentials.

11.2. Executive Milestones

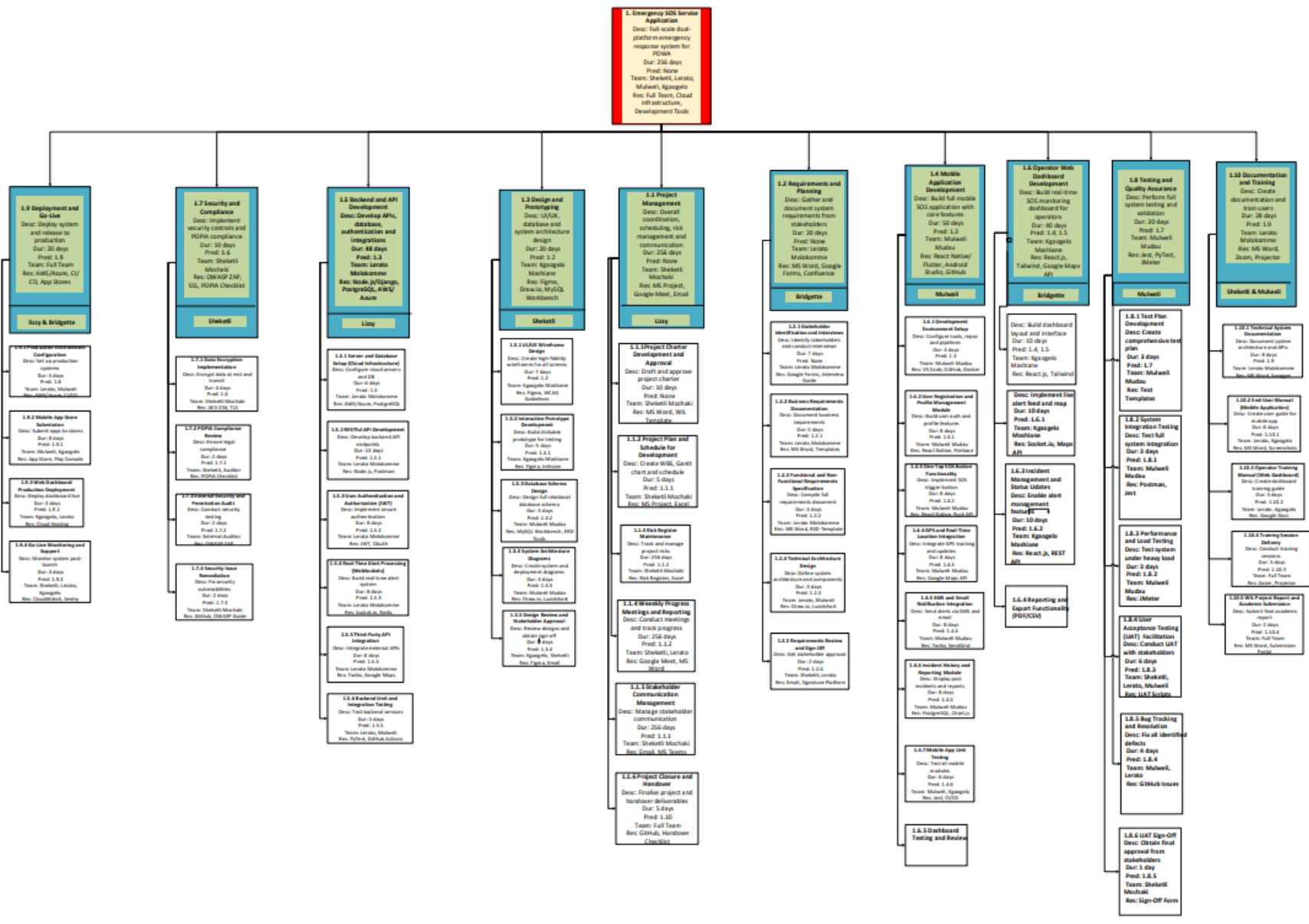
Milestone	Key Deliverable	Target Date	Responsible
Project Initiation Complete	Approved Project Charter signed by all parties	20 Mar 2026	Project Manager
Requirements Finalised	Requirements Specification Document (RSD) approved by POWA and supervisor	24 Apr 2026	Business Analyst
Design Approved	Architecture, UI/UX, and database design documents formally signed off	05 Jun 2026	Lead Developer / UI Designer
Sprint 1 Complete	Core SOS alert and GPS tracking functionality operational (Alpha)	03 Jul 2026	Full Developer Team
Sprint 5 Complete	Mobile app and web dashboard feature-complete (Beta)	28 Aug 2026	Full Developer Team
Development Phase Complete	Fully integrated E.SOS App prototype delivered for testing	06 Nov 2026	Full Developer Team
Testing Complete	All test reports passed; UAT sign-off received from POWA	31 Dec 2026	Quality Assurance Engineer
System Deployed	E.SOS App live in production environment	23 Jan 2027	Full Team
Project Closed	Final report, documentation, and formal handover to POWA complete	12 Feb 2027	Project Manager

11.3. Milestone Dependencies

Each executive milestone (Section 11.2, above) depends on the successful completion of preceding milestones and, in several cases, on external stakeholder input from POWA. The table below makes these dependencies explicit so that scheduling risk can be tracked against the critical path.

Milestone	Depends On	Target Date
Project Initiation Complete	POWA partnership agreement signed; team roles assigned	20 Mar 2026
Requirements Finalised	Project Initiation Complete; POWA stakeholder interviews and UAT-planning sessions	24 Apr 2026
Design Approved	Requirements Finalised; RSD sign-off by POWA and supervisor	05 Jun 2026
Sprint 1 Complete (Alpha)	Design Approved; development environment and CI/CD pipeline set up	03 Jul 2026
Sprint 5 Complete (Beta)	Sprint 1 Complete; stable API contracts and database schema	28 Aug 2026
Development Phase Complete	Sprint 5 Complete; all planned sprints executed without unresolved blocking defects	06 Nov 2026
Testing Complete	Development Phase Complete; test environment and UAT participants (POWA staff/end users) available	31 Dec 2026
System Deployed	Testing Complete; production cloud environment provisioned; app store submissions approved	23 Jan 2027
Project Closed	System Deployed; POWA operator training delivered; final documentation approved	12 Feb 2027

12. Work Breakdown Structure (WBS)

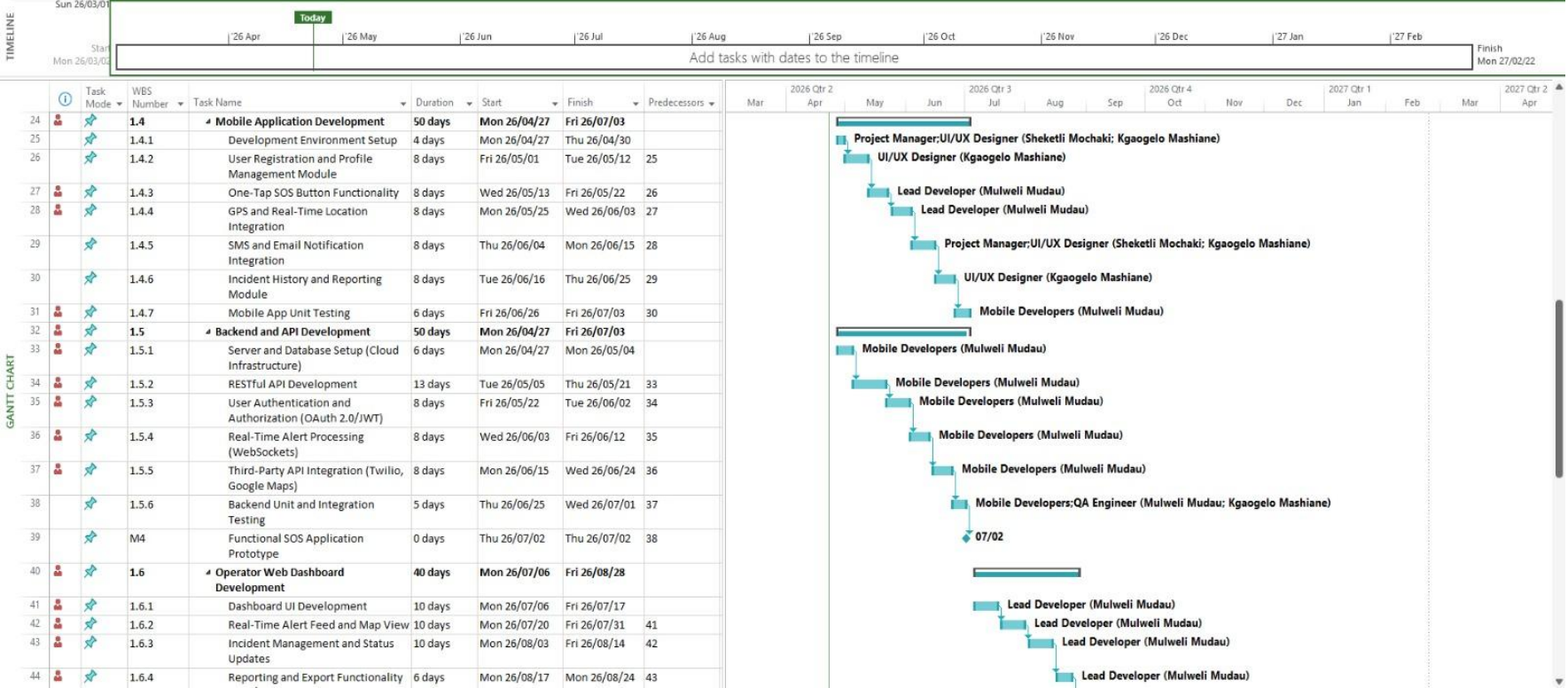
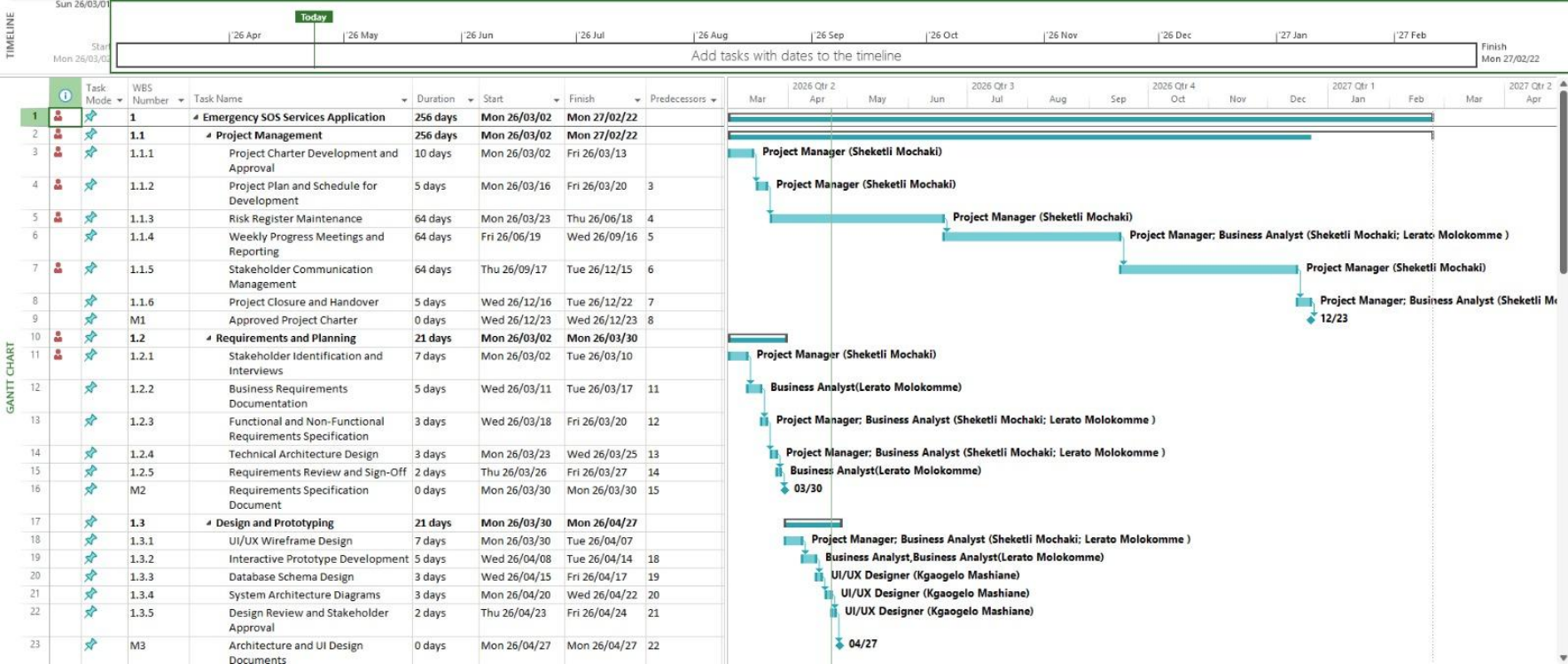


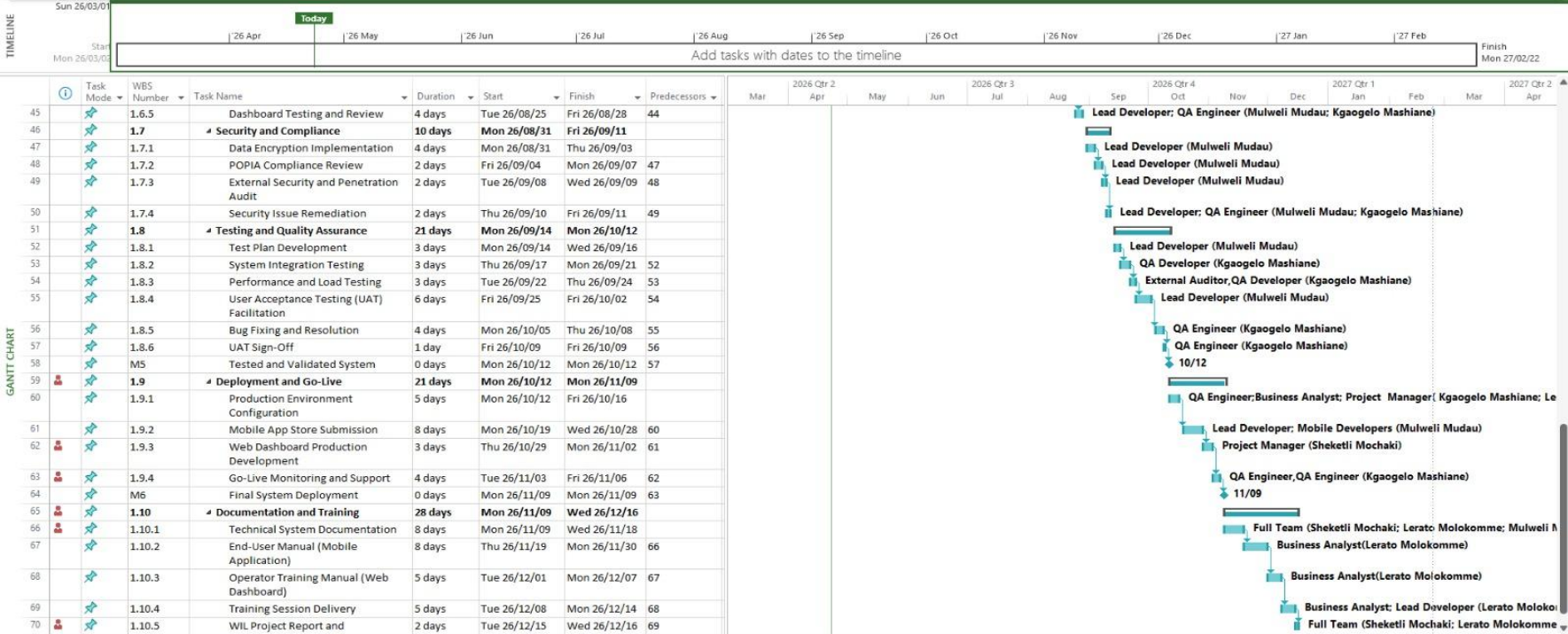
13. PROJECT SCHEDULES

13.1. Gantt chart and interpretation

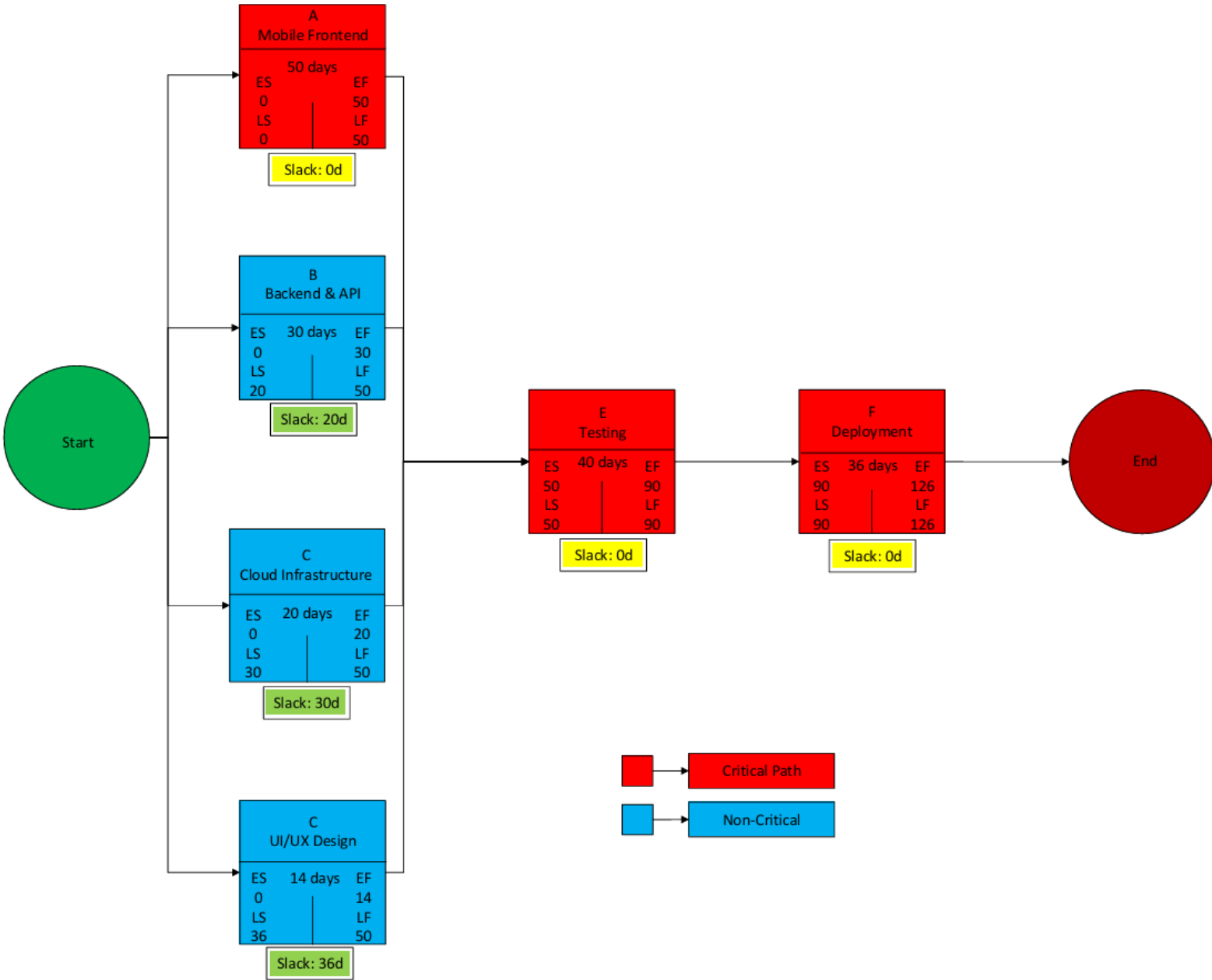
A Gantt chart is a horizontal bar chart that visually represents the project schedule, mapping each task against the calendar timeline. It allows the project manager and stakeholders to see briefly which tasks are active at any given point, the dependencies between tasks, and overall project progress. The Gantt chart for the E.SOS App project is presented in the Microsoft Project file submitted alongside this charter (MS Project E.SOS App Gantt chart Documentum).

The following table provides a textual Gantt summary across the full 256-day project lifecycle (02 March 2026 – 12 February 2027):





13.2. Pert chart and interpretation



14. Budget estimate

14.1. Funding source

The **POWA Project** is an academic Work Integrated Learning (WIL) project and therefore does not currently have a formal external funding source at this stage. The budget estimate provided below represents the realistic commercial cost that would be required if the project were to be fully developed and implemented in a professional environment. It is expected that **POWA (People Opposing Women Abuse)**, as the beneficiary non-governmental organisation, would seek funding from different external sources to support the full deployment and long-term running of the system. These sources may include **government grants**, especially from the **Department of Sport, Arts and Culture (DSAC)**, which provides financial support for social development and community-based initiatives.

Since POWA is a registered Non-Profit Organisation under the **NPO Act 71 of 1997**, it would qualify for this type of funding. Additional funding may also come from the **National Lottery Commission (NLC)**, which supports organisations involved in social development and gender-based violence prevention programmes. Furthermore, **Corporate Social Investment (CSI)** funding from major South African companies such as Absa Group Limited, Standard Bank Group, and MTN Group could be used to support the technological and operational costs of the project. International organisations such as UN Women, United States Agency for International Development (USAID), and the Bill & Melinda Gates Foundation may also provide funding support because these organisations often invest in projects that focus on women’s safety and gender-based violence response systems. Using a combination of these funding sources would help ensure that the project is financially sustainable and capable of making a long-term impact in the community (South African Government,1997).

14.2. Estimate

The following budget estimate is based on prevailing South African market rates for software development services, cloud infrastructure, and project management. All figures are expressed in South African Rand (ZAR).

#	Budget Item	Unit Cost (ZAR)	Quantity	Total Cost (ZAR)
1	Project Management (256 days / ~51 weeks)	R8,000/month	12.8 months	R102,400
2	Mobile App Development – React Native / Flutter	R25,000	1 app (dual platform)	R25,000
3	Web Dashboard Development – React.js	R20,000	1 dashboard	R20,000
4	Backend API Development – Node.js	R20,000	1 backend system	R20,000
5	UI/UX Design (Wireframes and Prototypes)	R12,000	1 project	R12,000
6	Database Design and Implementation	R8,000	1 database	R8,000
7	Cloud Infrastructure (AWS / Azure – 12 months)	R2,500/month	12 months	R30,000
8	Third-Party API Costs (Google Maps, Twilio, Firebase)	R5,000	12 months	R5,000
9	Security Audit and Penetration Testing	R15,000	1 audit	R15,000
10	Quality Assurance and Testing	R10,000	1 project	R10,000
11	User Training and Documentation	R8,000	1 session/package	R8,000
12	Domain Registration and SSL Certificates	R2,500	1 year	R2,500
13	Software Licences (development tools)	R5,000	1 year	R5,000
14	Contingency Reserve (10%)	—	—	R28,190

TOTAL PROJECT ESTIMATE	R 309 990
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The total estimate project is rounded to R320,000 including additional management buffer and unforeseen expenses. The extended 256-day timeline significantly increases project management and cloud infrastructure costs compared to a shorter engagement. The Work Integrated project is executed at zero direct cost to POWA, with student labour and institutional infrastructure serving as in-kind contributions.

15.Assumptions, Constraints and Risks

15.1. Assumptions

The following assumptions have been made in the planning of the Emergency SOS Services Project. These assumptions form the basis upon which the project plan, schedule, and budget have been constructed. If any assumption proves to be incorrect, a formal change request will be initiated.

Assumption	Rationale
POWA will provide timely access to relevant stakeholders for interviews and UAT sessions.	Requirements quality and UAT sign-off are entirely dependent on POWA's participation.
All team members will maintain consistent availability for the full 15-week project duration.	Team attrition or unavailability will impact the project schedule and quality.
Free-tier cloud infrastructure (AWS / Azure student accounts) will be sufficient for the development and testing phases.	Commercial cloud accounts are cost-prohibitive for a WIL project.
Third-party APIs (Google Maps, Twilio, Firebase) will remain available and functionally stable throughout the project.	API deprecation or breaking changes would require re-engineering.
The academic supervisor will provide feedback within 5 working days of each submission.	Timely academic feedback is needed to proceed between project phases.
Team members have the necessary technical skills or will acquire them within the first two weeks.	Skill gaps are expected to be resolved through self-learning and peer mentoring.
South African emergency service APIs are accessible for integration, subject to partnership agreements.	Integration scope may change if API access is not granted.

15.2. Constraints

Constraint Type	Constraint Type
Time	The project must be completed within 15 weeks as mandated by the WIL academic calendar. Extensions are not available without formal academic approval.
Budget	The project must be executed within the estimated R250,000 budget envelope. No additional funding mechanisms are available during the WIL phase.
Resources	The project team consists of exactly four students. No additional developers can be onboarded without academic and project manager approval.
Technology	The application must be compatible with Android 10+ and iOS 14+ devices and must function on mobile networks (3G minimum) given the target user base.

Legal / Compliance	All data handling must comply with the Protection of Personal Information Act (POPIA) and POWA's internal data governance policies.
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15.3. Risks Management

Risk	Mitigation Strategy	Contingency Plan
Scope Creep	Establish a formal change control process. All scope changes must be approved by the PM and POWA in writing before implementation.	Re-prioritise using MoSCoW method; defer lower-priority features to a Phase 2 release.
Cybersecurity Breach	Implement end-to-end encryption, OWASP security guidelines, regular penetration testing, and principle of least privilege access.	Activate incident response plan; notify affected users per POPIA requirements; engage security consultant.
Third-Party API Failure	Identify and document alternative API providers during the design phase. Implement abstraction layers to facilitate API switching.	Switch to backup API provider; revert to SMS-only emergency notification as fallback.
Team Member Unavailability	Crosstrain all team members in each other's primary responsibilities. Maintain detailed task documentation.	Redistribute tasks among remaining team members; request academic extension if impact is significant.
Budget Overrun	Track expenditure weekly against the budget. Use open-source tools wherever possible.	Activate the 10% contingency reserve; reduce scope of non-critical features.
Poor User Adoption	Involve POWA social workers and end users in the design and UAT phases. Conduct usability testing throughout development.	Conduct post-deployment training; redesign problematic UI elements based on user feedback.
Emergency Services Integration Failure	Begin partnership negotiations with SAPS and EMS in Phase 1. Treat integration as a 'Could Have' requirement.	Deploy without direct emergency service integration; rely on hotline directory and manual dispatch.
Performance Degradation	Conduct load testing with JMeter/k6 during Phase 5. Design the backend with horizontal scalability in mind.	Scale up cloud infrastructure; implement message queuing to manage concurrent alert spikes.
POPIA Non-Compliance	Engage a POPIA compliance checklist during the design phase. Implement data minimisation, consent management, and access controls.	Conduct immediate data audit; remediate identified gaps; document and report as required by law.

15.4. Risks

The table below outlines the risks to the POWA project, their probability and impact ratings, and the proactive steps to be taken to manage each risk:

#	Risk	Probability	Impact	Proactive Mitigation Steps
1	Team member leaves or drops out	Medium	High	Document all roles from inception; cross-train members on each other's tasks; maintain a shared drive so no knowledge is sold in one person.
2	No cooperation from sponsor / client	Medium	Very High	Sign a formal MOU or partnership agreement early; schedule monthly stakeholder check-ins; escalate to faculty advisor if engagement drops.
3	Survivors unwilling to participate	Medium	High	Partner with trusted GBV counsellors; guarantee full anonymity; use trauma-informed, non-coercive engagement methods approved by an ethics committee.
4	Team member does not contribute equally	Medium	Medium	Define individual deliverables in a RACI matrix; hold weekly stand-ups; address under-performance early through direct conversation and mediation if needed.
5	Running behind schedule	High	Medium	Build buffer weeks into the Gantt chart; track milestones weekly; apply MoSCoW prioritization to re-scope rather than compromise quality if delays occur.
6	Insufficient funding / budget overrun	Medium	High	Include a 10-15% contingency reserve; pursue multiple funding sources (NGOs, government grants, CSR); obtain written cost estimates before committing spend.
7	Lack of technical expertise	Low	Medium	Conduct a skills-gap assessment at kick-off; arrange targeted training or bring in a technical advisor; use accessible tools that lower the technical barrier.
8	Data privacy / confidentiality breach	Low	Very High	Obtain ethics clearance before data collection; anonymize all survivor data; enforce password-protected access; ensure full POPIA compliance for all team members.
9	Community resistance or reputational risk	Low	High	Run community sensitization workshops before launch; involve respected local advocates; prepare a communications plan to address misinformation swiftly.
10	Scope creep	High	Medium	Define and sign off a clear scope statement at inception; use a formal change-request process for any additions; review scope alignment at every milestone.
11	Poor team communication	Medium	Medium	Agree on a single communication platform; set expected response-time norms; hold fortnightly retrospectives to surface and resolve issues early.
12	Safety risk during fieldwork	Low	Very High	Never conduct fieldwork alone; implement a buddy check-in system; brief team on area-specific safety protocols; suspend fieldwork if security conditions deteriorate.

Risk Level Key: Very Low (minimal effect) | Low (Minor Disruption) | Medium (Moderate Effect) | High (Significant Impact) | Very High (Project-Threat)

16. Technical Feasibility Issues and Economic Feasibility Issues

16.1. Technical Feasibility Issues

The current hardware we are going to use is:	- HUAWEI MATEBOOK D16 i5 - HP ELITE WINDOWS 11 - LENOVO IDEA PAD - SAMSUNG GALXY A54(ANDRIOD 14) - IPHONE 12 PRO (IOS 26) - REAL TIME GPS TRACK ING ACCURACY - OFFLINE SOS FUNCTION - WINDOWS 11 OPERATING SYSTEM
Availability of resources:	80% AVAILABILTY
RESOLUTION / MIGITATION:	IMPLENTING A MUTLI-SOUCE LOCATION TRIANGUALTION USING GPS, WIFI AND CELLUAR NETWORK DATA. USING GOOGLE MAPS PLATFORM FOR GEOLOCATION API FOR ENCHANCED ACCURACY. IMPLEMENT An SMS-BASED SOS FALLBACK USING TWILIO SMS API, ENSURING ALREATS CAN BE SENT VIA CELLUAR NETWORK WIHTOUT INTERNAT CONNECTION.
The software we are going to use:	- ANDROID STUDIO - GOOGLE MAPS - TWILIO - FIREBASE - ORACLE - GITHUB - AWS FREE TIER STUDENT ACCOUNTS - GOOGLE PLAY - IOS APP STORE
Availability of resources:	80% AVAILABILTY
RESOLUTION/MIGITATION	IMPLEMENT AES-256 ENCRYPTION FOR DATA AT REST, TLS 1.3 FOR DATA IN TRANSIT, JWT-BASED AWS ELASTIC BEANSTALK OR AZURE APP SERVIES FOR AUTO-SCALING USE OF DOCKERS CONTAINERS TO STANDERISE THE DEVELOPMENT ENVIROMENT

16.2. Economic Feasibility Issues

ACTIVITY	TEAM MENBER	TIME ALLOCATED	RAND/HOUR
1.1 PROJECTT MAMANAGEMNT	SHAEKETLI MOCHAKI	212	200
1.1.1 PROJECT CHARTER DEVELOPMENT	SHAKETHI MOCHAKI	240	200
1.1.2 PROJECT CHATER AND SHECDULE DEVELOPMENT	SHAKETHI MOCKAKI	120	200

1.1.2 RISK RESGISTER MAINTAINACE	SHAKETHI MOCKAKI	720	200
1.1.4 WEEKLY PROGRESS AND REPORTING	SHAKETHI MOCKAKI	1864	200
1.1.5 STAKEHOLDER COMMUNICATION MANAGMENT	SHAKETHI MOCKAKI	28	200
1.1.6 PROJECT CLOSURE AND HANDOVER	SHAKETHI MOCKAKI	120	200
1.2 REQUIREMNTS AND PLANNING	LERATO MOLOKOMME	480	156
1.2.1 STAKEHOLDER IDENTIFICATION	LERATO MOLOKOMME	168	156
1.2..2 GET STAKEHOLDER APPROVAL	LERATO MOLOKOMME	120	156
1.2.3 FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS	LERATO MOLOKOMME	72	156
1.2.4 TECHNICAL ARCHITECTURE DESIGN	LERATO ,MULWELI	72	156
1.2.5 REQUIREMNTS REVIEW AND SIGN –OFF	SHEKETHI , LERATO	48	156
1.3 DESIGN AND PROTOYING	KGAOGELO MASHINE	480	236
1.3.1 DESIGN AND PROTOYPING	KGAOGELO MASHIANE	70	236
1.3.2 INTERACTIVE PROTOTYPING DEVELOPMENT	KGAOGELO MASHIANE	50	236
1.3.3 DATABASE SCHEMA DESIGN	MULWELI MUDAU	30	236
1.3.4 SYSTEM ARCHITECURE DIAGRAMS	MULWELI MDUAU	30	236
1.3.5 DESIGN REVIEW AND STAKEHOLDER APPROVAL	SHEKETHLI	20	236
1.4 MOBILE APPLICATION DEVELOPMENT	MULWELI MUDAU	1200	546
1.4.1 DEVELOPMENT ENVIROMENT SETUP	MULWELI MUDAU	40	546
1.4.2 USER REGISTRATION AND PROFILE MANAGMENT MODULE	MULWELI MUDAU	80	546
1.4.3 ONE TOP SOS BUTTON FUNCUALITY	MULWELI MUDAU	80	546
1.4.4 GPS AND REAL-TIME LOCATION	MULWELI MUDAU	80	546
1.4.5 SMS AND EMAIL NOTIFIACION INTERGATION	MULWELI MUDAU	80	546

1.4.6 INCIDENT HISTORY AND REPORTING MODULE	MULWELI MUDAU	80	546
1.4.7 MOBILE APP UNIT TESTING	MULWELI MUDAU KGAOGELO MASHIANE	80	546
1.5 BACKEND AND API DEVELOPMENT	LERATO MOLOKOMME	1152	200
1.5.1 SERVER AND DATABASE SETUP	LERATO MOLOKOMME	60	200
1.5.2 RESTFUL API DEVELOPMET	LERATO MOLOKOMME	130	200
1.5.3 USER AUTHENTICATION AND AUTHORAZATION	LERATO MOLOKOMME	8 0	200
1.5.4 REAL-TIME ALRET PROCESSING	LERATO MOLOKOMME	450	200
1.5.5 THIRD-PARTY API INTERGRATION	LERATO MOLOKOMME	230	200
1.5.6 BACKEND UNIT AND INTERGRATION TESTING	LERATO MOLOKOMME MULWELI	430	200
1.6 OPERATOR OWEB DASHBOARD DEVELOPMENT	KGOAGELO MASHIANE	960	400
1.7 SECURITY AND COMPLIANCE	SHEKETLI MOCHAKI	240	400
1.8 TESTING AND QUALITY ASSURANCE	MULWELI MUDAU	480	400
1.9 DEPLOYMENT AND GO-LIVE	FULL TEAM	480	130
1.9.1 PRODUCTION ENVIROMENT CONFIGURATION	FULL TEAM	80	130
1.9,2 MOBILE APP STORE SUBMISSION	FULL TEAM	76	130
1.9.3 WEB DASHBOARD PRODUCTION DEVELOPMENT	FULL TEAM	120	130
1.9.4GO-LIVE MONITORING AND SUPPORT	FULL TEAM	400	130
1.10 DECOMENTING AND TRAINING	LERATO MOLOKOMME	672	376
1.10.1 TECHNICAL SYSTEM DOCUMENTATION	LERATO MOLOKOMME	110	376
1.10.2 END-USER MANUAL	LERATO MOLOKOMME KGAOGELO	78	376
1.10.3OPERATOR TRAINING MANUAL	LERATO MOLOKOMME KGAOGELO	74	376
1.10.4 TRAINING SESSION DELIVERY	FULL TEAM	56	376

1.10.5 WIL PROJECT CHARTER REPORT AND ACADEMIC SUBMISSION	FULL TEAM	56	376
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17. Project Organization

17.1. Roles and Responsibilities

The E.SOS App project team consists of four students from the IIE Rosebank College Information Technology in Software Development programme. Each team member holds a primary role with defined responsibilities, while also contributing to cross-functional activities as needed. The RACI matrix below defines each role's level of involvement (R = Responsible, A = Accountable, C = Consulted, I = Informed).

Responsibility Area	Sheketli Mochaki (PM)	Lerato Molokomme (BA)	Mulweli Mudau (Lead Dev)	Kgaogelo Mashiane (UI/QA)	POWA (Client)
Project Planning and Scheduling	A/R	C	C	C	I
Stakeholder Communication	A/R	R	I	I	C
Requirements Gathering	A	R	C	C	R
System Architecture Design	I	C	A/R	C	I
Database Design	I	I	A/R	C	I
UI/UX Design	C	C	C	A/R	C
Mobile App Development	C	R	R	C	I
Web Dashboard Development	I	I	C	A/R	I
Backend / API Development	I	C	A/R	C	I
Security Implementation	I	C	A/R	C	I
Testing and QA	C	C	C	A/R	C
Documentation	A	R	C	C	I
POWA Liaison	A/R	R	I	I	C
Budget Tracking	A/R	I	I	I	I
Project Closure / Handover	A/R	R	R	R	R

Key: R = Responsible | A = Accountable | C = Consulted | I = Informed

17.2. Stakeholders (Internal and External)

Stakeholder management is critical to the success of the E.SOS App project. The following tables identify all internal and external stakeholders, their interests, and their level of engagement.

Internal Stakeholders

Stakeholder	Role	Interest / Expectation	Engagement Strategy
Sheketli Mochaki	Project Manager	Successful project delivery, WIL academic credit, professional portfolio development.	Daily stand-ups, weekly progress reports, milestone tracking.
Lerato Molokomme	Business Analyst	Accurate requirements capture, client satisfaction, academic achievement.	Stakeholder interviews, documentation reviews, UAT facilitation.
Mulweli Mudau	Lead Developer	Technical solution quality, code review standards, deployment success.	Sprint planning, code reviews, technical design sessions.
Kgaogelo Mashiane	UI Designer / QA Engineer	Accessible, intuitive design; comprehensive test coverage; quality assurance.	Design workshops, usability testing sessions, test planning.

External Stakeholders

Stakeholder	Role	Interest / Expectation	Engagement Strategy
POWA (NGO Client)	Project Beneficiary / Client	A reliable, accessible emergency SOS system that enhances service delivery to GBV survivors.	Regular requirement reviews, UAT sessions, progress presentations.
POWA Social Workers	Primary System Operators	An easy-to-use dashboard for real-time alert monitoring and coordinating crisis responses.	Involvement in UAT; training sessions during deployment phase.
GBV Survivors / End Users	Primary Mobile App Users	A discreet, intuitive app that provides immediate help in a crisis.	Representative user testing during UAT; accessibility testing.
South African Police Service (SAPS)	Emergency Response Partner	Accurate, real-time incident data to enable faster dispatch.	Partnership negotiation letters; API integration documentation.
Tshwane Emergency Services (EMS)	Emergency Response Partner	Timely, accurate location data for medical emergency dispatch.	Integration planning meetings; API specification sharing.
Department of Social Development	Regulatory Stakeholder	Compliance with social development regulations and POPIA data protection laws.	Compliance documentation; POPIA compliance report.
Cloud/API Providers (AWS, Google, Twilio)	Technology Partners	Proper usage of their services within licensing and rate limit agreements.	Service agreement review; API usage monitoring dashboards.

18. **Project Charter Approval**

The undersigned acknowledge that they have reviewed the Emergency SOS Services Project Charter and formally authorise the project team to proceed with the design, development, and deployment of the E.SOS App for POWA (People Opposing Women Abuse). By signing below, each party confirms their understanding of and commitment to the project's objectives, scope, timeline, budget, and governance structure as documented in this charter. Any material changes to the project scope, budget, or timeline will require a formal change request, reviewed and approved by all undersigned parties or their designated representatives

Role:	Project Manager	Date:	23/04/2026
Signature:	S.M.M	Student No:	ST10452756
Full Name:	Sheketli Mochaki	Title:	Mr.

Role:	POWA Executive Director	Date:	23/04/2026
Signature:	F.S	Title:	Ms
Full Name:	Fatima Shaik		

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Declaration of Authenticity

I, __Sheketli Mochaki__ ID Number, __0201115056081__ hereby declare that this portfolio, and any evidence included therein, contains my own independent work and that I have not received help from other groups.

I confirm that we have not committed plagiarism in the accomplishment of this work, nor have I falsified and/or invented experimental data.

I accept the academic penalties that may be imposed for violations of the above.

STUDENT NUMBER: __ST10452756__

STUDENT NAME: __Sheketli Mochaki__

STUDENT SIGNATURE: __S.M.M__ DATE: __23/04/2026__

I, _____Lerato Molokomme_____ ID Number, _____080805253083_____ hereby declare that this portfolio, and any evidence included therein, contains my own independent work and that I have not received help from other groups.

I confirm that we have not committed plagiarism in the accomplishment of this work, nor have I falsified and/or invented experimental data.

I accept the academic penalties that may be imposed for violations of the above.

STUDENT NUMBER: _____ST104396226_____

STUDENT NAME: _____Lerato Molokommme_____

STUDENT SIGNATURE: _____L.M_____ DATE: _____23/04/2026_____

I, _____Mulweli Mudau_____ ID Number, _____0504030736087_____ hereby declare that this portfolio, and any evidence included therein, contains my own independent work and that I have not received help from other groups.

I confirm that we have not committed plagiarism in the accomplishment of this work, nor have I falsified and/or invented experimental data.

I accept the academic penalties that may be imposed for violations of the above.

STUDENT NUMBER: _____ST10358894_____

STUDENT NAME: _____Mulweli Mudau_____

STUDENT SIGNATURE: _____M.M_____ DATE: _____23/04/2026_____

I, _____Kgaogelo Mashiane _____ ID Number, _____ hereby declare that this portfolio, and any evidence included therein, contains my own independent work and that I have not received help from other groups.

I confirm that we have not committed plagiarism in the accomplishment of this work, nor have I falsified and/or invented experimental data.

I accept the academic penalties that may be imposed for violations of the above.

STUDENT NUMBER: _____ST10446936_____

STUDENT NAME: _____Kgaogelo Mashiane _____

STUDENT SIGNATURE: _____K.M_____ DATE: _____23/04/2026_____

Peer Evaluation

(NOT TO BE SHARED WITH TEAM MEMBERS)

Please complete and include the peer-evaluation reports when submitting your final PoE.

GROUP NAME/NUMBER: Group 5 DATE: 23/04/2026

Please rate each of your team members using the details in the assessment criteria descriptions provided. Marks received from each member of the team will be added up and then averaged for each member's individual mark. Complete a separate evaluation form for each team member.

Name of student being evaluated:		Never	Seldom	Frequently	Always
Kgaogelo Mashiane		0	1	2	3
The student's personal work					
1.	The student contributed good ideas that added value to the project.				3
2.	The student performed their tasks in line with what was expected of them.				3

3.	The student produced high quality work.				3
4.	The student managed their own time well and met deadlines.				3
5.	The student connected their own learning and skills to a real-world problem.				3
The student's work as part of a team (when relevant)					
6.	The student accepted responsibility for a fair portion of the tasks.				3
7.	The student was an enthusiastic member of my team.				3
8.	The student helped others to be successful.				3
9.	The student worked well with other members of the team.				3
10.	The student showcased respect and dignity towards all members of the team.				3
TOTAL:					30/30

Comments: ____

Excellent work and great contribution to the team.

NAME: ____Mulweli Mudau_____ STUDENT NUMBER: _____ST10358894_____

SIGNATURE: ____M.M_____

(NOT TO BE SHARED WITH TEAM MEMBERS)

Please complete and include the peer-evaluation reports when submitting your final PoE.

GROUP NAME/NUMBER: __Group 5_____ DATE: _____23/04/2026_____

Please rate each of your team members using the details in the assessment criteria descriptions provided. Marks received from each member of the team will be added up and then averaged for each member's individual mark. Complete a separate evaluation form for each team member.

Name of student being evaluated:	Never	Seldom	Frequently	Always
Lerato Molokomme				
	0	1	2	3
The student's personal work				

1.	The student contributed good ideas that added value to the project.				3
2.	The student performed their tasks in line with what was expected of them.				3
3.	The student produced high quality work.				3
4.	The student managed their own time well and met deadlines.				3
5.	The student connected their own learning and skills to a real-world problem.				3
The student's work as part of a team (when relevant)					
6.	The student accepted responsibility for a fair portion of the tasks.				3
7.	The student was an enthusiastic member of my team.				3
8.	The student helped others to be successful.				3
9.	The student worked well with other members of the team.				3
10.	The student showcased respect and dignity towards all members of the team.				3
TOTAL:					30/30

Comments: Excellent work and great contribution to the team.

NAME: Sheketli Mochaki STUDENT NUMBER: ST10452756

SIGNATURE: S.M.M

(NOT TO BE SHARED WITH TEAM MEMBERS)

Please complete and include the peer-evaluation reports when submitting your final PoE.

GROUP NAME/NUMBER: Group 5 DATE: 23/04/2026

Please rate each of your team members using the details in the assessment criteria descriptions provided. Marks received from each member of the team will be added up and then averaged for each member's individual mark. Complete a separate evaluation form for each team member.

Name of student being evaluated:	Never	Seldom	Frequently	Always
Sheketli Mochaki	0	1	2	3

The student's personal work					
1.	The student contributed good ideas that added value to the project.				3
2.	The student performed their tasks in line with what was expected of them.				3
3.	The student produced high quality work.				3
4.	The student managed their own time well and met deadlines.				3
5.	The student connected their own learning and skills to a real-world problem.				3
The student's work as part of a team (when relevant)					
6.	The student accepted responsibility for a fair portion of the tasks.				3
7.	The student was an enthusiastic member of my team.				3
8.	The student helped others to be successful.				3
9.	The student worked well with other members of the team.				3
10.	The student showcased respect and dignity towards all members of the team.				3
TOTAL:					30/30

Comments:

NAME: _____ Lerato Molokomme _____ STUDENT NUMBER: _____ ST10439626 _____

SIGNATURE: _____ L.M. _____

(NOT TO BE SHARED WITH TEAM MEMBERS)

Please complete and include the peer-evaluation reports when submitting your final PoE.

GROUP NAME/NUMBER: ____ Group 5 _____ DATE: ____ 23/04/2026 _____

Please rate each of your team members using the details in the assessment criteria descriptions provided. Marks received from each member of the team will be added up and then averaged for each member's individual mark. Complete a separate evaluation form for each team member.

Name of student being evaluated:	Never	Seldom	Frequently	Always
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Mulweli Mudau		0	1	2	3
The student's personal work					
1.	The student contributed good ideas that added value to the project.				3
2.	The student performed their tasks in line with what was expected of them.				3
3.	The student produced high quality work.				3
4.	The student managed their own time well and met deadlines.				3
5.	The student connected their own learning and skills to a real-world problem.				3
The student's work as part of a team (when relevant)					
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8.	The student helped others to be successful.				3
9.	The student worked well with other members of the team.				3
10.	The student showcased respect and dignity towards all members of the team.				3
TOTAL:		30/30			

Comments: Excellent work and great contribution to the team.

NAME: _____ Kgaogelo Mashiane _____ STUDENT NUMBER: _____

SIGNATURE: _____ K.M. _____

Self-Reflection Report

Mudau Mulweli

ST10358894

Introduction

Work Integrated Learning (WIL) plays an important role in bridging the gap between academic knowledge and real-world application. Through this project, I was able to apply theoretical concepts to a practical scenario, which helped me gain a deeper understanding of how systems are designed, planned, and implemented. The WIL project also exposed me to teamwork, problem-solving, and professional communication, all of which are essential skills in the IT and business environment. Overall, this experience helped me grow both academically and personally.

Skills Learnt

During this WIL project, I developed and applied several important skills across different areas:

Industry-Specific Skills

I improved my **report writing and documentation skills**, especially when working on structured sections like the project overview, budget estimates, and system requirements. I also learned how to organise information professionally and follow academic formats such as Harvard referencing. Additionally, I gained experience in **planning and structuring a system**, including understanding how features, users, and stakeholders interact.

Interpersonal Communication Skills

Working in a group helped me develop my **communication skills**, especially during discussions, brainstorming sessions, and decision-making. I learned how to listen to others' ideas, share my own input, and provide feedback respectfully. Group meetings required clear communication to ensure everyone understood their tasks and responsibilities.

Management Skills

I developed **time management skills** by meeting deadlines and balancing different parts of the project. I also improved my ability to handle pressure when tasks were due at the same time. In situations where problems occurred, such as confusion or delays, I applied basic **problem-solving and crisis management skills** to find solutions and keep the work progressing.

Role in the Team

During the WIL project, I worked as part of a team where each member had specific responsibilities. I contributed mainly to sections such as the **Project and Product Overview** and **Budget Estimate, Project plan** ensuring that my work aligned with the rest of the group document.

Leadership and Instructions

Although I was not the main leader, I followed instructions given by the group leader and contributed ideas during discussions. I also took responsibility for completing my assigned tasks on time and ensuring quality work.

Contribution to Team Success

My contribution helped ensure that our document was complete, well-structured, and met the required academic standards. I made sure my sections were detailed, clear, and aligned with the project objectives.

Group Dynamic

The group dynamic was generally positive. We worked together, shared ideas, and supported each other when needed. Everyone had different strengths, which helped improve the overall quality of the project.

Dealing with Concerns and Conflict

There were moments where misunderstandings or differences in opinions occurred. These were handled through discussion and compromise. We focused on the goal of completing the project successfully rather than personal disagreements.

Research, Technology and Presentation of Information

During this project, I was required to find relevant and useful information to complete my tasks.

Scenario 1

I had to research information about **POWA (People Opposing Women Abuse)** to understand their services and how the system could support their work.

- **Where I found information:** Official POWA website and online articles
- **Technology used:** Google search, web browser
- **Presentation:** I used this information in written paragraphs within the report using proper academic structure and referencing

Scenario 2

I had to research **budget estimates and system features** for the application.

- **Where I found information:** Online resources, examples of similar systems, and academic materials
- **Technology used:** Internet research tools and Microsoft Word
- **Presentation:** Information was presented in both paragraph form and tables (e.g., budget table and feature table)

This process helped me understand the importance of selecting reliable sources and presenting information clearly.

Personal Strengths and Weaknesses

Tasks I Did Well

- Writing structured and detailed sections of the report
- Researching relevant information for the project
- Following academic formatting and referencing
- Meeting deadlines for my assigned tasks
- Contributing ideas during group discussions

My Strengths

1. Good research skills
2. Ability to organise and present information clearly
3. Strong written communication
4. Ability to work well in a team
5. Time management and meeting deadlines

Tasks I Did Not Do Well

- Sometimes I took longer to complete tasks due to overthinking
- I was not always confident when sharing ideas in the group

Reasons for Challenges

These challenges were mainly due to a lack of confidence and wanting my work to be perfect before submitting it.

How I Can Improve

- Improve confidence by participating more in discussions
- Manage time better by avoiding overthinking tasks
- Practice faster decision-making when working under pressure

Stakeholder Relationship

During the project, the lecturer acted as a guide and evaluator. The relationship was professional, and the lecturer provided clear instructions and feedback when needed. This helped the team stay on track and understand the expectations of the project.

Within the team, relationships were cooperative and respectful. Communication was important in ensuring that everyone contributed and understood their roles. Overall, the teamwork experience helped me understand the importance of collaboration in real-world projects.

Sheketli Mochaki

ST10452756

Introduction

Work Integrated Learning (WIL) gave me the opportunity to move from studying software development concepts in theory to applying them on a live, socially significant project. As Lead Developer on the Emergency SOS Services Application (E.SOS App), developed in partnership with POWA (People Opposing Women Abuse), I was responsible for the technical backbone of the project across both the planning stage and the requirement analysis and system design stage: assessing technical feasibility, defining the system architecture, designing the database, and embedding security into the platform from the ground up.

Working on a safety-critical application for survivors of gender-based violence sharpened my appreciation for the weight that technical decisions can carry. A poor architecture choice or an unverified assumption in this kind of system is not just a bug — it could directly affect someone's safety. That responsibility shaped how I approached every technical task across the project, from the earliest feasibility assessment in the project charter through to the detailed database and security design in the system design document.

Skills Learnt

Industry-Specific Skills

During the charter phase, I contributed to the Technical Feasibility section, evaluating hardware and software availability, and proposing mitigation strategies such as multi-source location triangulation (GPS, Wi-Fi, and cellular data) and an SMS-based fallback using the Twilio API for offline SOS functionality. This required me to think beyond 'what works in theory' and consider real deployment constraints — device compatibility, network reliability in under-resourced areas, and the cost implications of cloud infrastructure choices. I also contributed technical architecture input to the Work Breakdown Structure, helping define the backend, API, and cloud infrastructure work packages, and assisted in identifying technical risks such as third-party API failure and performance degradation under concurrent load, both of which fed into the project's risk register.

In the requirement analysis and system design phase, I translated the use cases into a class diagram, defining the User, SOSAlert, IncidentReport, EmergencyContact, and POWAOperator classes along with their relationships (composition, aggregation, association, and dependency). This required me to think carefully about ownership and lifecycle — for example, deciding that an SOSAlert is a composition of User (it cannot exist without an owning user) while EmergencyContact is an aggregation (a contact can exist independently of any specific alert event). I also led the database design, producing the normalised table structures for users, sos_alerts, incident_reports, emergency_contacts, and messages, and the accompanying entity relationship diagram. Designing for POPIA compliance from the outset — deciding which fields needed encryption at rest, applying data minimisation, and defining retention-relevant timestamps — gave me hands-on experience applying data protection law to a real schema rather than just reading about it. On the architecture side, I helped define the three-tier MVC structure (React Native/React.js presentation layer, Node.js/Express application layer, Firebase Firestore and PostgreSQL data layer) and justified each technology choice against the project's actual constraints, such as needing offline-capable real-time updates (Firestore) alongside relational integrity for audit logs (PostgreSQL).

Interpersonal Communication Skills

Translating technical constraints into language the rest of the team and our client could understand was a skill I had to actively develop. Explaining why certain technology choices mattered — for example, why an SMS fallback was non-negotiable for a GBV support tool —

required me to communicate technical reasoning in terms of user impact rather than implementation detail. Presenting the class diagram and ERD to the team later in the project required me to explain object-oriented and relational modelling concepts to teammates with different technical backgrounds. I learned to walk through diagrams incrementally — starting with the core entities before introducing relationship types — so that design reviews were genuinely collaborative rather than a one-way technical briefing.

Management Skills

I managed my own technical workstream independently across both phases: scoping the feasibility assessment, prioritising which technical risks needed mitigation strategies before development could begin, and later sequencing the dependencies between my design tasks and the work of the UI/QA Engineer and Business Analyst, since the database schema and API contracts needed to be stable before wireframing and interaction-flow design could be finalised. This taught me to plan technical deliverables deliberately, flagging which design decisions were blocking other people's work rather than letting bottlenecks surface late.

Role in the Team

Leadership and Responsibilities

As Lead Developer, I held primary responsibility (Accountable/Responsible in the RACI matrix) for System Architecture Design, Database Design, Backend/API Development, and Security Implementation throughout the project. During the charter phase this meant assessing whether the team's proposed technology stack, hardware, and third-party services (Google Maps, Twilio, Firebase, AWS) were realistic given our skill levels, budget, and timeline, and flagging technical risks early rather than leaving them as open concerns. In the system design phase, this translated into concrete artefacts: the high-level architecture table, the technology stack with justifications, the security architecture (TLS 1.3, AES-256, JWT with refresh rotation, RBAC, OWASP-aligned input validation), the class diagram, and the full database schema and ER diagram.

Contribution to Team Success

By flagging feasibility issues and proposing mitigations during the charter stage, rather than discovering them mid-development, I helped the team avoid the kind of late-stage technical surprises that derail project timelines. Later, by producing a stable, requirement-traceable architecture and schema, I gave the rest of the team a firm technical foundation to design around. The UI/QA Engineer's wireframes and the Business Analyst's use-case documentation could both reference concrete data structures rather than placeholders, which kept the project's technical and design work internally consistent.

Group Dynamic

The team worked cooperatively, with each member's specialisation respected. As the team member with the deepest technical background, I was often consulted on feasibility questions raised by other sections of the project, such as budget estimates for backend and cloud services, and design reviews later on were collaborative: feedback from the Business Analyst on use case completeness directly shaped which entities and relationships I included in the class diagram, and accessibility requirements from the UI/QA Engineer influenced how I structured user-facing data fields, such as language preference and notification settings.

Dealing with Concerns and Conflict

There were occasional disagreements about the appropriate level of technical detail and direction — for example, whether to treat SAPS system integration as guaranteed functionality (I advocated for treating it as a 'Could Have' given uncertain partnership timelines), and later whether to use a single database technology or the hybrid Firestore/PostgreSQL approach we ultimately adopted. I argued for the hybrid model because real-time alert data and structured audit/compliance data have genuinely different access patterns. Resolving both disagreements through a clear comparison of trade-offs, rather than personal preference, was a useful exercise in grounded technical decision-making.

Research, Technology and Presentation of Information

Scenario 1: Technology Stack and Feasibility Research

- Where I found information: official documentation for Google Maps Platform, Twilio, Firebase, and AWS; comparative articles on React Native vs Flutter for cross-platform development.
- Technology used: Google, official API documentation, GitHub, Android Studio.

- Presentation: Findings were distilled into the Technical Feasibility Issues table, pairing each technical risk with a concrete resolution/mitigation strategy rather than a general statement.

Scenario 2: Architecture, Security and Database Research

- Where I found information: OWASP Top 10 documentation, POPIA legislation, database design textbooks, UML class diagram conventions, and official PostgreSQL/Firestore documentation.
- Technology used: Microsoft Word, Microsoft Project, diagramming software, internet research tools.
- Presentation: Information was structured into the WBS technical streams and risk register as tables, and later into the Security Architecture section and database schema/ERD, with visual diagrams used wherever a technical audience needed a precise, scannable specification rather than narrative text.

This process reinforced the importance of citing authoritative technical sources (official API and legislative documentation over forum posts) when making architecture and data-handling decisions that the whole team and the client would rely on.

Personal Strengths and Weaknesses

Tasks I Did Well

- Assessing technical feasibility realistically rather than optimistically
- Designing a normalised, POPIA-aware database schema
- Producing a class diagram with correctly justified relationship types
- Selecting and justifying a technology stack against real project constraints
- Embedding security requirements directly into the architecture rather than treating them as an afterthought

My Strengths

1. Strong technical problem-solving and systems thinking
2. Security-conscious design thinking
3. Ability to justify technical decisions with clear reasoning
4. Comfortable researching and validating third-party tools before committing to them
5. Effective at sequencing technical dependencies for the team

Tasks I Did Not Do Well

- I sometimes focused too narrowly on technical correctness and under-estimated how long it would take to explain those decisions to non-technical stakeholders
- I initially under-documented some of the rationale behind class relationship choices, which meant I had to revisit and clarify them during review
- I could have involved the rest of the team earlier in reviewing the database schema, rather than presenting a more finished version

Reasons for Challenges

These challenges stemmed from genuinely wanting the technical plan and design to be sound before sharing them, which meant I sometimes spent longer than planned validating assumptions or working through artefacts independently before bringing them to the team.

How I Can Improve

- Timebox feasibility and design research more tightly and flag open questions rather than resolving everything alone before moving on
- Share draft technical artefacts earlier, even when incomplete, to get feedback sooner
- Document design rationale alongside diagrams themselves, not as a separate afterthought
- Continue building confidence in presenting technical trade-offs persuasively to non-technical stakeholders

Stakeholder Relationship

My key stakeholders throughout the project were my own teammates, who depended on a stable feasibility assessment, architecture, and schema to plan and design their own sections, and indirectly POWA and its operators, whose operational needs — reliability, offline access, low-cost sustainability, and survivor confidentiality — shaped every technical decision I made. I made sure technical recommendations were explained in terms of what they meant for POWA's day-to-day use of the system, and kept every design decision traceable back to the documented functional and non-functional requirements, so the system we designed would genuinely serve POWA's needs rather than just being technically elegant.

Conclusion

Working as Lead Developer on the E.SOS App project taught me that technical leadership starts well before any code is written — it begins with honest feasibility assessment and carries through into every architecture, data, and security decision that follows. Designing systems intended to support survivors of gender-based violence gave real weight to decisions I might otherwise have treated as purely technical, such as offline fallback support, data protection compliance, and user safety. This experience has strengthened my systems design ability, my security awareness, and my confidence in justifying technical decisions to a broader team, and has prepared me to take on more senior technical leadership responsibility in future professional environments.

Lizzy Molokomme — Lead Web Developer & UI/UX

St10439636

Main Role:

Lead Web Developer & UI/UX Designer

Primary Responsibility:

Responsible for designing and developing the web-facing components of the E.SOS system while ensuring that the interface is intuitive, accessible, responsive, secure, and suitable for users who may be experiencing an emergency.

Key Responsibilities

Web Development

- Develop the POWA E.SOS web interface.
- Implement responsive web pages for desktop, tablet, and mobile devices.
- Develop the operator-facing web dashboard.
- Implement navigation between major system sections.
- Integrate frontend components with backend APIs.
- Display real-time SOS alerts and incident information.
- Implement user authentication and role-based access.
- Integrate emergency and support functionality into the interface.
- Ensure reliable communication between the frontend and backend.

UI/UX Design

- Design the overall user interface and user experience.
- Create wireframes and high-fidelity interface designs.
- Design the SOS activation experience around **speed, simplicity and accessibility**.
- Design clear navigation and information hierarchy.
- Design emergency alert screens and confirmation messages.
- Design operator dashboard screens for monitoring SOS incidents.
- Design forms for user information, emergency contacts and incident reporting.
- Apply accessibility considerations for users with different abilities.
- Consider multi-language support within the interface.
- Conduct design reviews and improve interfaces based on feedback.

Technical Collaboration

- Work with the Lead Developer/technical team to ensure UI designs are technically feasible.
- Work with the Business Analyst to ensure interfaces satisfy documented requirements.
- Work with QA to identify usability and interface issues.
- Ensure the frontend follows the agreed system architecture and API contracts.

Security & Privacy

- Ensure sensitive information is not unnecessarily exposed through the interface.
- Apply secure authentication practices.
- Design appropriate access levels for survivors, operators and administrators.
- Consider POPIA requirements when displaying and collecting personal information.

Main Deliverables

- Website UI
- Operator Dashboard UI
- Mobile/web interface designs
- Wireframes
- High-fidelity prototypes
- Navigation structure
- SOS alert interface
- Incident reporting interface
- User account interface
- Accessibility considerations
- Frontend implementation
- API-integrated web pages
- UI/UX testing and improvements

Kgoagelo Mashiane — Project Manager

St10446936

Main Role:
Project Manager

Primary Responsibility:

Responsible for coordinating the project from initiation through completion, ensuring that the team remains aligned with the project scope, requirements, schedule, resources, risks and stakeholder expectations.

Key Responsibilities**Project Planning**

- Develop and maintain the overall project plan.
- Define project objectives, scope and deliverables.
- Break the project into manageable work packages.
- Develop and maintain the Work Breakdown Structure (WBS).
- Establish project milestones and deadlines.
- Coordinate activities between different team members.
- Ensure dependencies between tasks are identified.

Team Coordination

- Allocate responsibilities to team members.
- Coordinate the work of the Lead Developer, UI/UX Developer, Business Analyst and QA team members.
- Organise regular team meetings.
- Track individual and team progress.
- Ensure team members understand their responsibilities.
- Encourage collaboration between technical and non-technical team members.
- Resolve issues that could prevent team members from completing their work.

Requirements & Scope Management

- Ensure project activities remain aligned with approved requirements.
- Monitor changes to project scope.
- Prevent unnecessary scope creep.
- Ensure functional and non-functional requirements are considered throughout the project.
- Coordinate requirement reviews with the Business Analyst and development team.

Risk Management

- Identify potential project risks.
- Maintain the project risk register.
- Assess the likelihood and impact of identified risks.
- Develop mitigation and contingency strategies.
- Monitor technical, resource, schedule and integration risks.
- Escalate significant risks to stakeholders when required.

For example, the project identified risks such as **third-party API failure and performance degradation under concurrent load**, which required mitigation planning.

Stakeholder Management

- Act as a communication link between the project team and stakeholders.
- Communicate project progress.
- Gather and communicate stakeholder feedback.
- Ensure stakeholder expectations are understood.
- Keep project decisions and requirements traceable.
- Ensure POWA's operational requirements remain central to project decisions.

The project documentation identifies teammates and POWA/operators as key stakeholders, with requirements including reliability, offline access, sustainability and survivor confidentiality.

Quality Management

- Monitor whether project deliverables meet agreed requirements.
- Coordinate testing activities.
- Ensure defects and issues are recorded and followed up.
- Coordinate User Acceptance Testing (UAT).
- Ensure project documentation is complete.
- Monitor compliance with project quality requirements.

Main Deliverables

- Project Charter
- Project Management Plan
- Work Breakdown Structure
- Project Schedule
- Milestone Plan
- Risk Register
- Responsibility/RACI Matrix
- Meeting Records
- Progress Reports
- Requirements Tracking
- Change Management Records
- Quality Management Activities
- Testing Coordination
- Final Project Documentation

The project deliverables include the **Project Charter, Requirements Specification, System Architecture Document, UI/UX prototypes, Mobile Application, Web Operator Dashboard, Security and Compliance Report, and Test Reports.**